

A Data- Driven Framework for The Selective Metal Leaching from Spent Lithium-Ion Battery Based on Small Dataset

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Abstract

Abbreviations Key

ANN – Artificial Neural Network

BAL – Bayesian Active Learning

DT – Decision Tree

GBR – Gradient Boosting Regression

KNN – K-Nearest Neighbors

Li – Lithium

Co – Cobalt

Mn – Manganese

Ni – Nickel

ML – Machine Learning

MLP – Multilayer Perceptron

RF – Random Forest

SVR – Support Vector Regression

SVM – Support Vector Machine
SHAP – SHapley Additive exPlanations
PDP – Partial Dependence Plot
RMSE – Root Mean Squared Error
MAE – Mean Absolute Error
 R^2 – Coefficient of Determination
EDA – Exploratory Data Analysis
LIB – Lithium-Ion Battery

1. Introduction

The rapid expansion of electric vehicles and grid-scale energy storage over the years has led to a rapid increase in the demand for lithium-ion batteries (LIBs), making end-of-life battery management both a resource security issue and an environmental necessity. More importantly, it is projected that LIBs market would grow tremendously up to ten times its current capacity by year 2030 [1]. SLIB contains cathode materials made of compounds such as Lithium iron phosphate (LiFePO_4) and $\text{LiNi}_x\text{Co}_y\text{Mn}_z\text{O}_2$ (NCM) together with AL collector [2]. For such a material, recycling or upcycling is particularly important because these materials contain strategically important metals including Li, Ni, Co, and Mn, whose supply chains remain vulnerable to geographic concentration and market volatility [3]. At the same time, mismanaged

spent LIBs can release hazardous constituents, create fire risks, and increase secondary lithium-ion batteries (SLIBs) a central challenge in building a circular battery economy.

Several recycling routes have been developed, each with distinct advantages and limitations. Physical and mechanical pretreatment methods such as crushing, sieving, magnetic separation, electrostatic separation, and delamination are attractive because they are comparatively simple and can reduce downstream chemical demand [4,5]. However, their separation efficiency is often limited by the strong adhesion among cathode active material, conductive carbon, PVDF binder, and Al current collector. As a result, incomplete liberation is common and this could lead to the production of mixed “black mass” streams with variable impurity levels and inconsistent composition. This variability becomes especially problematic for NCM materials, where downstream recovery performance depends strongly on how effectively Al foil, Cu, binder residues, and other inactive components are removed before chemical processing.

Pyrometallurgical recycling is industrially mature and tolerant of feed heterogeneity, which explains its continued commercial relevance [5,6]. It can efficiently concentrate transition metals into alloy phases and destroy organics and electrolytes. Yet pyrometallurgy is energy intensive, usually sacrifices lithium and aluminum to slag or secondary streams unless additional treatment is added, and often requires extensive downstream refining to recover individual metals at high purity [6,7]. These drawbacks reduce its attractiveness for selective recovery from NCM cathodes when the objective is to maximize resource efficiency and produce high-purity precursors for closed-loop reuse.

Among the available approaches, hydrometallurgy and especially leaching-based recovery remains the most versatile and practical pathway for SLIBs recycling. Leaching can operate at lower temperatures than pyrometallurgy, enables comparatively high metal recoveries, and offers the possibility of selective dissolution under controlled solution chemistry [8,9]. Acid systems based on sulfuric, hydrochloric, nitric, organic acids, or deep eutectic solvents have all been investigated [7–9], often combined with reducing agents to accelerate dissolution of transition metals. Because leaching transfers metals into a controllable liquid phase, it is also compatible with downstream solvent extraction, precipitation, and crystallization steps needed to produce battery-grade products. For these reasons, hydrometallurgical leaching remains the dominant framework in both research and industrial flowsheets for LIB recycling.

Still, saying leaching is the best available route does not mean it is straightforward. Its core limitation is that leaching performance depends on many tightly coupled variables, including acid concentration, reductant dosage, temperature, solid-to-liquid ratio, reaction time, cathode chemistry, particle size, binder residue, and impurity content. Under harsh conditions, valuable metals dissolve efficiently, but so do undesirable components such as Al, Cu, and Fe, which complicate purification and reduce product quality. Under milder conditions, selectivity may improve, but dissolution kinetics can become too slow or incomplete. Recent studies have shown that impurities are not passive spectators: Al, Cu, and Fe can alter reductant consumption, participate in side reactions, and change dissolution behavior in ways that are difficult to predict a priori [10]. Chernyaev et al.[10] for example, showed that Cu, Fe, and Al interfere with reductive sulfuric-acid leaching of NMC materials by affecting hydrogen peroxide consumption and side reactions, while newer work has further highlighted the overlooked role of aluminum impurities in shaping NCM dissolution behavior.

This complexity explains why selective leaching remains difficult to optimize using conventional one-factor-at-a-time or purely empirical trial-and-error experimentation. The design space is multidimensional, nonlinear, and chemistry-dependent, while experimental methods are expensive because each run consumes reagents, time, and valuable battery-derived material. Moreover, the available datasets in LIB recycling are often small, noisy, and heterogeneous, because experiments are generated across different battery chemistries, pretreatment histories, and analytical protocols [10–12]. Under these conditions, data-driven methods become especially valuable not because they replace mechanistic understanding, but because they can extract structure from sparse experimental data, identify the most influential process variables, and accelerate the search for selective operating windows with fewer experiments.

Recent studies support this transition toward data-driven optimization. Feng et al. [3] applied machine learning (ML) to metal leaching from spent LiFePO_4 materials and showed that data-driven models can capture the contributions of process and composition variables to recovery outcomes. Nogueira et al. [12] proposed an applied ML framework for LIB hydrometallurgical recycling. The framework connects metal leaching prediction with preliminary economic and environmental screening, showing how predictive models can support greener process development. More broadly, Bayesian optimization is particularly well suited to

chemical and materials problems characterized by small, noisy, and expensive datasets, because it uses surrogate models and acquisition strategies to guide the next most informative experiment instead of exhaustively sampling the full design space. That logic is directly relevant to selective metal leaching from spent LIBs, where each experiment is costly and the objective is not merely high dissolution, but selective dissolution under practical operating constraints.

Based on this context, a data-driven framework for selective metal leaching from spent LIBs is well justified. Existing recycling routes either lack sufficient selectivity, require high energy input, or remain too sensitive to feed purity for broad deployment. Leaching remains the most flexible recovery platform, yet its optimization is hindered by multivariate interactions and limited experimental data. A small-dataset data-driven approach can therefore fill an important gap by learning from sparse leaching experiments, identifying the dominant variables that govern selective dissolution, and guiding process design toward higher recovery efficiency, lower impurity co-dissolution, and reduced experimental burden. In that sense, data-driven selective leaching is not only a modeling exercise; it is a practical pathway toward more adaptive, resource-efficient, and scalable recycling of SLIBs. Therein lies the motivation for this study. Using the small dataset available in literature, the present study proposes a novel way to optimize the prediction of precious metals during SLIBs hydrometallurgical recycling.

2. Methodology

2.1 Data Collection and ML workflow.

The dataset used in this study was compiled from multiple peer-reviewed literature sources. A complete spreadsheet including each source is documented in supplementary information. The data includes information on experimental studies that applies various hydrometallurgical pathways for recovery Li, Co, Mn and Ni from SLIB. The compiled dataset consists of eight numeric process input variables including the lithium feed (SLIB) content (Li in feed %), cobalt feed content (Co in feed %), manganese feed content (Mn in feed %), nickel feed content (Ni in feed %), leaching agent concentration in mol/L, reducing agent concentration in %, leaching time

in minutes, and temperature in °C as well as two categorical process variables: the type of leaching agent and the type of reducing agent used. The target outputs were the leaching efficiencies of lithium (Li), cobalt (Co), manganese (Mn), and nickel (Ni), each expressed as a percentage.

Figure 1 illustrates the end-to-end pipeline from data collection and preprocessing, through the dual-pipeline model screening and hyperparameter tuning phase, feature importance analysis, Bayesian active learning augmentation, and final forward model development. Details of each workflow is described in subsequent sections.

2.2 Data Preprocessing and ML model selection

Raw data were imported from an Excel file and cleaned prior to model development. Target leaching efficiency values exceeding 100% were removed as physically inadmissible. Categorical features were standardized by stripping whitespace, applying title case, and replacing data records with missing leaching or reducing agent as "Unknown". Numeric target columns were cast to floating-point format prior to training.

A single preprocessing pipeline was constructed using scikit-learn's ColumnTransformer. Numeric features were processed through median imputation, while categorical variables were encoded using one-hot encoding with unknown-category handling. No standard scaling was applied. All preprocessing steps were embedded within the model pipeline to prevent data leakage.

In addition to the eight base numeric inputs, five interaction and polynomial features were engineered to capture nonlinear process relationships: the product of leaching time and temperature ($\text{Time} \times \text{Temp}$), the product of leaching agent concentration and time ($\text{Leach_Conc} \times \text{Time}$), the square of temperature (Temp^2), the square of time (Time^2), and the ratio of leaching agent concentration to reducing agent concentration (Acid_to_Reducer). This yielded a total of 13 numeric features alongside the two categorical variables.

The dataset was partitioned into training and testing subsets using an 80/20 split with a fixed random seed (42) to ensure reproducibility, consistent with conventions established in prior ML studies [15–18].

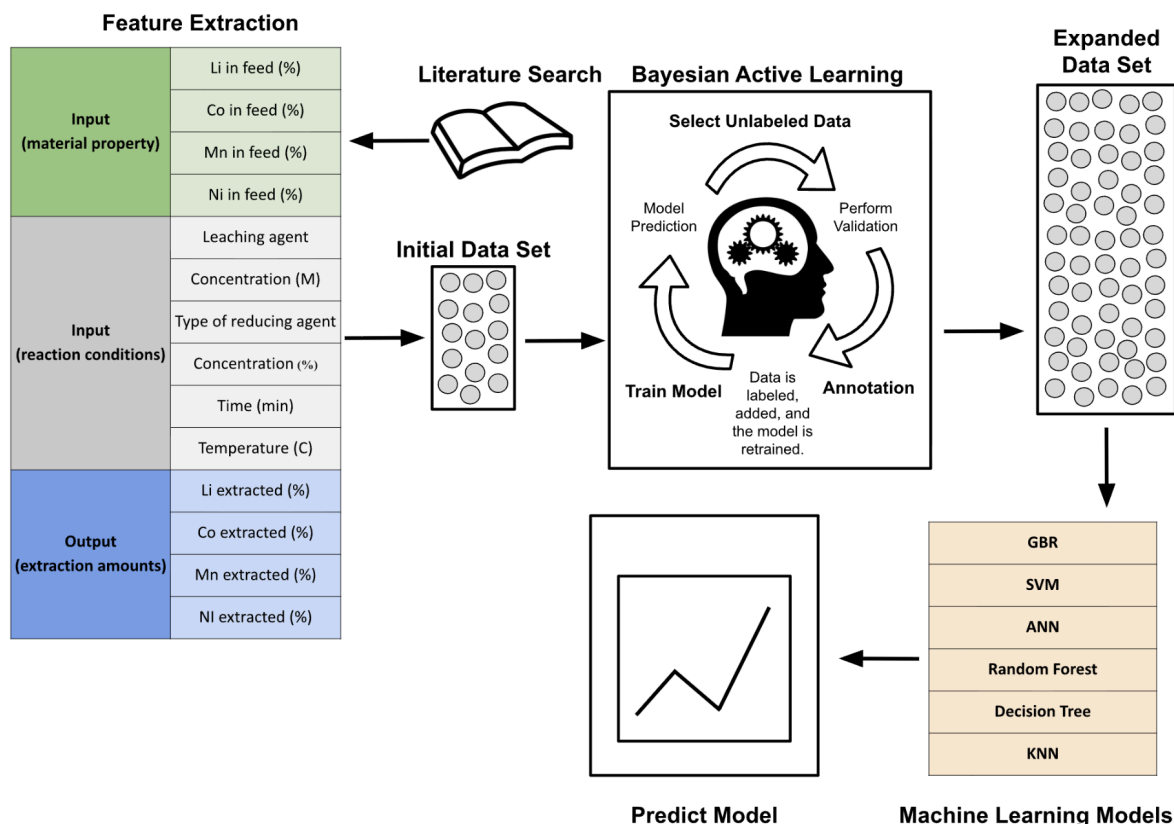


Figure 1. Workflow schematic of machine learning pipeline.

A single algorithm, namely Gradient Boosting Regression (GBR) implemented via scikit-learn's GradientBoostingRegressor, was selected for model development. A separate GBR model was trained for each of the four target leaching efficiencies (Li, Co, Mn, Ni), with training performed exclusively on rows where a valid (non-missing) value was available for the corresponding metal. Fixed hyperparameters were used across all four models: 300 estimators, maximum depth of 4, learning rate of 0.05, subsample ratio of 0.8, and a minimum of 5 samples per leaf. Model performance was assessed on the held-out test set (20%) using R^2 , MAE, and RMSE. Generalization was further evaluated through 5-fold cross-validation, with R^2 and MAE reported as mean $\pm$ standard deviation across folds.

2.3 Feature Analysis and Removal of Categorical Variables

Feature importance was computed for each trained model to assess the contribution of individual input variables to model predictions. For tree-based models (RF, GBR, DT), SHAP values were calculated using the TreeExplainer from the SHAP library, which provides theoretically grounded, additive attribution scores based on Shapley values from cooperative game theory. The mean absolute SHAP value was computed across all samples to produce a global importance ranking for each feature. In cases where SHAP was unavailable or failed, permutation importance was used as a fallback, in which each feature's values are randomly shuffled and the resulting drop in model performance is used as a proxy for importance. Following feature importance analysis, the variables with least contributions to the model were removed from the feature set, and all candidate models were retrained using only the eight continuous numeric inputs.

2.4 Bayesian Active Learning

To address data sparsity and improve the representativeness of the training dataset, a synthetic data augmentation procedure was implemented in place of the Bayesian Active Learning (BAL) framework. The augmentation strategy leverages a GBR model trained on the original dataset to generate physically plausible synthetic samples that preserve the statistical characteristics and group structure of the compiled data.

First, a GBR model was trained on the original dataset for each target metal, using the eight base numeric features and the two categorical variables as inputs. Calibrated, group-specific noise levels were estimated from the residuals of condition-grouped subsets within each leaching agent and reducing agent combination, using the median absolute deviation of within-group residuals as the noise standard deviation (minimum 0.5%). New process conditions were then generated by resampling observed conditions within each group with replacement, followed by the addition of Gaussian perturbations scaled to 5% of the observed range for each condition variable. Reducing agent concentration was held at zero for groups in which no reducing agent was used. GBR model predictions were generated for each synthetic condition set and augmented with the calibrated group-specific noise. All synthetic target values were clipped to the physically admissible range of [0, 100%] and additionally constrained to within ± 5 percentage points of the

observed range for each metal within the corresponding group, preserving the NaN structure of metals not reported for a given group.

The augmented dataset was generated at two scales: a standalone synthetic dataset targeting approximately 5,000 total rows with group sizes proportional to the original data, and a combined dataset in which each group was augmented by a factor of five relative to its original size, then merged with the original data. The latter combined dataset was used as input for final model training. The statistical fidelity of the synthetic data was validated by comparing means, standard deviations, interquartile ranges, and out-of-bounds rates between the original and synthetic distributions, as well as verifying that synthetic condition ranges did not exceed those observed in the original data.

3. Results and Discussion

3.1 Exploratory Data Analysis

The summary statistics for the compiled combined dataset are presented in Table 1. The dataset contains 992 entries across the process variables. The feed compositions range as follows: Co content ranges from 0.10% to 60.20%, reflecting the diversity of NMC and LCO materials represented in the literature; Ni content ranges from 0.00% to 36.00%, and Mn content from 0.00% to 19.53%. Leaching agent concentrations range from 0.10 to 6.00 mol/L, while reducing agent concentrations range from 0.00% to 25.00%, and leaching times vary between 0 and 1,080 minutes at temperatures between 20°C and 100°C.

The target leaching efficiencies show strong central tendencies: median Li recovery is 88.09%, median Co recovery is 77.69%, median Mn recovery is 85.94%, and median Ni recovery is 86.58%, indicating that the majority of reported experiments achieve high extraction yields. The relatively high standard deviations, particularly for Mn (26.74%) and Co (23.55%), reflect the breadth of experimental conditions captured across source studies, reinforcing the need for robust preprocessing and imputation strategies. Despite the heterogeneous nature of the data, standardization to common units and consistent imputation of missing values are performed to ensure that the dataset is suitable for comparative machine learning analysis.

Table 1. Summary statistics of original dataset used. Note concentration (M) represents the leaching agent concentration while concentration in % represents reducing agent concentration.

	count	mean	std	min	25%	50%	75%	max	median
Li in feed %	990	4.40	2.24	0.00	2.87	4.40	6.50	7.42	4.40
Co in feed %	990	28.72	16.97	0.10	16.00	24.73	46.06	60.20	24.73
Mn in feed %	990	4.78	6.21	0.00	0.14	2.00	7.81	19.53	2.00
Ni in feed %	990	8.94	10.68	0.00	0.30	2.69	17.13	36.00	2.69
Concentration, M	990	1.84	1.26	0.10	1.00	1.50	2.00	6.00	1.50
Concentration %	990	1.71	2.94	0.00	0.00	0.80	2.00	25.00	0.80
Time, min	990	80.28	122.72	0.00	30.00	60.00	100.00	1080.00	60.00
Temperature, C	990	65.22	18.52	20.00	50.00	70.00	80.00	100.00	70.00

Li	917	82.82	22.56	0.00	76.00	92.00	99.00	100.0 0	92.00
Co	948	69.78	26.93	0.00	49.75	78.09	93.00	100.0 0	78.09
Mn	353	77.43	27.59	0.00	60.00	90.00	100.0 0	100.0 0	90.00
Ni	406	77.85	25.02	10.00	65.00	90.00	98.00	100.0 0	90.00

A Pearson correlation heatmap of the dataset is shown in Figure 2. Among the process inputs, temperature and leaching agent concentration show moderate positive correlations with all four leaching efficiencies. For instance, temperature shows positive correlations with Li (0.25), Co (0.35), Ni (0.45), and a slight negative correlation with Mn (-0.12). Reagent concentration (M) shows positive correlations with Li (0.18), Co (0.13), Mn (0.46), and Ni (0.30). The results are generally consistent with the established hydrometallurgical principles in which elevated temperature and higher acid concentration generally enhance metal dissolution kinetics. For instance, Musariri et al. [13] study showed that an increase in temperature led to an elevation in leaching rate for Co, Li, and Ni leaching with citric acid and malic acid.

Reducing agent concentration showed a clear positive association with Co and Mn leaching efficiency, indicating its important role in promoting the dissolution of transition-metal oxides. Reagent concentration (%) shows a correlation of 0.26 with Co extraction, 0.17 with Mn extraction, 0.23 with Li extraction, and 0.21 with Ni extraction. In acid leaching of SLIBs battery cathodes, reductants are commonly used to convert less soluble higher-valence species into more

soluble lower-valence forms, thereby accelerating dissolution kinetics and increasing metal recovery. This effect is especially important for Co-bearing cathodes, where Co^{3+} in layered oxides is reduced to the more soluble Co^{2+} state. Some studies of LIB hydrometallurgy consistently identify hydrogen peroxide as the most widely used reductant for this purpose, while also noting that other organic and inorganic reductants can play a similar role in enhancing transition-metal extraction [14]. The stronger correlation observed for Co is particularly well supported. Zhao et al. [14] reported that adding ethanol as a reducing agent during sulfuric-acid leaching of spent LiCoO_2 increased both the dissolution rate and extraction efficiency by reducing Co^{3+} to Co^{2+} ; under optimized conditions, both Li and Co recoveries exceeded 99%. Their study directly links reductant presence to improved cobalt leaching, rather than merely to acid availability.

A similar explanation can be extended to Mn-containing cathode systems. In mixed Ni–Co–Mn cathode materials, reductive conditions facilitate the breakdown of manganese-containing oxide lattices and improve Mn release into solution, especially when Mn is present in less soluble oxidized states. Recent work using sugar-based reductants achieved >85% Mn leaching together with >88% Co and Ni and >98% Li, confirming that reductive leaching can simultaneously improve recovery of multiple transition metals from cathode materials [15].

Additionally, there are strong positive correlations between the extraction efficiencies of the different metals (e.g., Co and Ni at 0.83; Li and Co at 0.75; Li and Mn at 0.73). This indicates that the dissolution of these metals is highly interdependent. Because Li, Ni, Co, and Mn share the same structural matrix in layered NMC cathodes, the chemical breakdown of the bulk oxide lattice inherently forces the simultaneous release of all constituent metals.

Apart from that, there is a notable negative correlation (around -0.40) that exists between temperature and leaching time. This can be connected to the standard experimental design in hydrometallurgy, where higher temperatures provide the activation energy needed to accelerate reaction kinetics, allowing for significantly shorter processing times. Interestingly, leaching time exhibits slight negative correlations with the target extractions (e.g., -0.28 for Ni, -0.25 for Mn). This suggests that simply extending the leaching duration without adequate temperature or acid

concentration does not improve yields and may even allow for secondary precipitation or the formation of passivation layers on the unreacted particle cores.

The heatmap also reveals the chemical stability of high-Mn structures, where Mn Feed (%) shows a distinct negative correlation (-0.35) with Mn Extraction (%). This is because as the relative proportion of manganese in the cathode material increases, it becomes proportionally more difficult to achieve high extraction efficiencies for it due to robust, insoluble MnO_2 phases during acid leaching that causes reaction bottleneck

Lastly, the correlation between Reagent Concentration (M) and Reagent Concentration (%) is near zero (-0.02) which might imply isolated effects of acidity versus reductive potential in the leaching experiment.

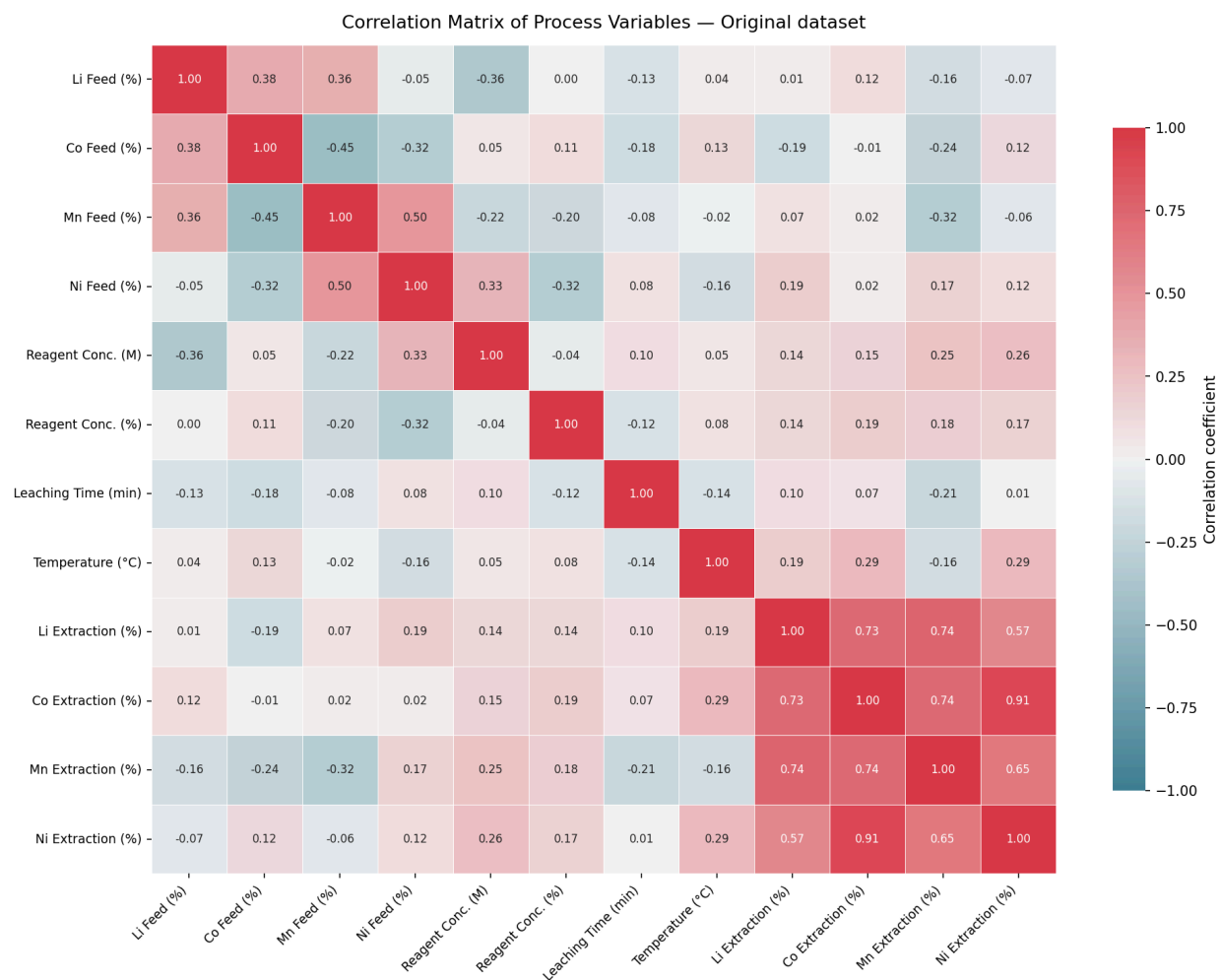


Figure 2. Heat map — Original dataset.

Figure 2. Heat map of the dataset.

The strong positive correlations observed among Li, Co, Mn, and Ni extraction efficiencies (ranging from 0.60 to 0.83) in the combined dataset reinforce the conclusion that a shared dissolution environment governs the recovery of all four metals. Once the complex cathode matrix is sufficiently destabilized by acid attack and reductive conversion, the simultaneous release of these transition metals and lithium becomes highly favored. This aligns with the premise that leaching efficiency is predominantly controlled by the thermodynamic and kinetic conditions set by the process variables, rather than the initial feed composition. Indeed, the initial feed percentages exhibit very weak correlations with overall extraction efficiencies (e.g., Li feed % correlates with Li extraction at just 0.03). Instead, process conditions act as the primary drivers: acid concentration (M) shows a notable positive influence on Mn (0.46) and Ni (0.30)

extraction, while the reducing agent (%) uniformly promotes the dissolution of all targeted metals (0.17 to 0.26). Interestingly, the expanded dataset reveals more complex kinetic behaviors across broader experimental conditions; while elevated temperatures strongly favor Ni (0.45) and Co (0.35) extraction, extended leaching times exhibit inverse relationships with Mn (-0.25) and Ni (-0.28) recovery, suggesting that unnecessarily prolonged leaching may lead to secondary precipitation or competing side reactions.

3.2 Initial Model Screening (With Categorical Variables)

The machine learning evaluation results for the Gradient Boosting Regressor (GBR) on the original dataset are summarized in Table 2. These metrics reflect 5-fold cross-validation performance rather than a single train-test split, providing a more robust and conservative estimate of model generalization. The GBR achieved training R^2 values of 0.9521, 0.9360, 0.9535, and 0.9469 for Li, Co, Mn, and Ni respectively, indicating strong in-sample fit consistency across all folds. On the validation folds, however, test R^2 values fell to 0.8080, 0.7903, 0.7558, and 0.7613 for Li, Co, Mn, and Ni respectively, with corresponding test MAEs of 6.07, 8.09, 7.20, and 7.16 percentage points.

This training-to-validation gap of approximately 15–18 percentage points implies the presence of mild overfitting when model is trained on a relatively small and heterogeneous dataset. The engineered interaction and polynomial features increase the effective dimensionality of the input space, meaning the model partially learns the training distribution rather than fully generalizable kinetic relationships when data volume is limited. Lithium achieved the highest and most stable validation R^2 among all four targets (0.8080), consistent with its comparatively homogeneous recovery behavior across conditions. Manganese exhibited the lowest validation R^2 (0.7558) alongside the highest test RMSE (13.19%), reflecting the greater sensitivity of Mn dissolution to undersampled combinations of cathode composition and reagent conditions.

Table 2. R^2 , RMSE and MAE for the original dataset (with categoricals)

Base Model Metrics	Model	Train R²	Train RMSE	Train MAE	Test R²	Test RMSE	Test MAE
Li extracted (%)	GBR	0.9521	4.93	3.40	0.8080	9.80	6.07
Co extracted (%)	GBR	0.9360	6.81	4.71	0.7903	12.23	8.09
Mn extracted (%)	GBR	0.9535	5.90	3.19	0.7558	13.19	7.20
Ni extracted (%)	GBR	0.9469	5.72	3.47	0.7613	11.78	7.16

The parity plots in Figure 3 below provide a clear visual confirmation of the Gradient Boosting Regressor's fit, illustrating the distribution of predicted versus actual leaching efficiencies for each target metal on the combined (original + augmented) dataset.

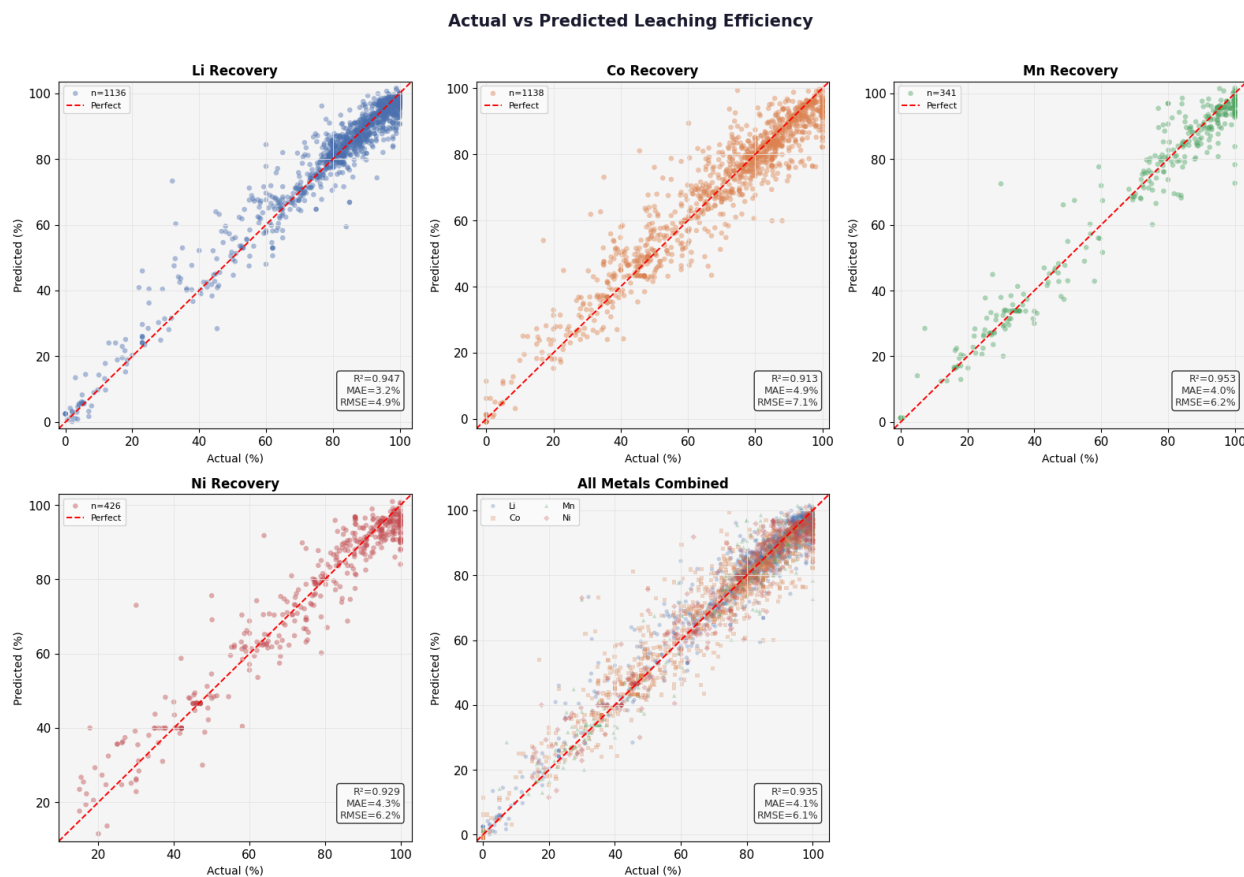


Figure 3. Forward parity plots for Li, Co, Mn, and Ni (Combined dataset) after 5 cross-validation folds

The parity plots demonstrate a strong linear correlation between the experimental and predicted values, with the vast majority of data points clustered tightly along the $y=x$ reference line across the entire response range. Notably, the models exhibit excellent predictive accuracy in the high-efficiency region (80%–100%), which is the most critical operational domain for industrial hydrometallurgical processes seeking maximum metal recovery. For instance, the Lithium (Li) plot shows an overwhelmingly dense, tight cluster bounded between 80% and 100% on both axes, visually corroborating its exceptional test R^2 (0.9634) and remarkably low test MAE (2.50). Cobalt (Co) similarly shows dense packing in the 70% to 100% range with minimal deviation from the ideal fit line.

While accuracy is exceptionally high at the upper end of the scale, there is more noticeable scatter observed at lower extraction efficiencies, particularly for Manganese (Mn) and Nickel

(Ni) below the 60% mark. For example, in the Mn and Ni plots, actual experimental values between 10% and 40% exhibit wider vertical spread, with predictions sometimes deviating by $\pm 10\%$ to 15% from the $y=x$ line. This increased dispersion at lower ranges can be largely attributed to the inherent distribution of the literature data used for training; because published studies primarily report optimized conditions, the dataset is heavily skewed toward high extraction yields. Consequently, the GBR model is exposed to a much higher density of training examples in the 80%–100% regime. It is highly probable that if a broader supply of experimental data representing unoptimized, lower-efficiency conditions were available, the model's predictive accuracy would improve in these regions, and this low-end scattering would be significantly reduced. Nevertheless, the tight clustering across the overall operational range visually corroborates the high R^2 and low RMSE metrics, confirming that the GBR architecture is highly suitable for capturing the complex, multi-variable interactions governing the acid leaching of spent lithium-ion battery cathodes.

3.3 Feature Importance and Removal of Categorical Variables

SHAP-based feature importance rankings for all four leaching efficiency targets are presented in Figure 4, where the results show predictor hierarchies across metals, reflecting differences in the underlying dissolution mechanisms governing each target.

For Co and Ni recovery, temperature emerged as the dominant or co-dominant predictor, with mean $|\text{SHAP}|$ values of 3.427 and 5.135 respectively. This can be explained by the role of temperature in enhancing leaching kinetics through increased reaction rates, improved mass transfer, and accelerated solid–liquid interfacial reactions. Particularly in acid leaching systems, elevated temperature reduces activation energy barriers and promotes the breakdown of cathode crystal structures, which in turn facilitates the release of metal ions into solution. Numerous studies on spent lithium-ion battery leaching have reported significant improvements in metal recovery with increasing temperature, particularly for Co and Ni dissolution from layered oxide cathodes [17–19].

For Li recovery, feed composition variables dominated, with Co Feed (%) and Ni Feed (%) ranking first and second with mean |SHAP| values of 3.864 and 3.065 respectively, followed by Temperature (°C) at 2.678. This suggests that Li dissolution is strongly co-governed by the compositional context of the cathode matrix, likely reflecting the structural interdependencies between Li sites and transition metal occupancy in layered oxide phases. Leaching agent concentration and reducing agent concentration also contributed meaningfully to Li recovery (2.175 and 2.024), indicating that process conditions retain secondary but non-negligible influence.

Mn recovery exhibited the most distinct pattern among all targets, with Mn Feed (%) and Leaching Agent Concentration dominating by a substantial margin (8.986 and 8.073), dwarfing all other features. This indicates that manganese dissolution is predominantly governed by the intrinsic properties of the cathode material and reagent availability rather than by thermal or temporal process conditions. This behaviour is supported by prior studies showing that Mn can exist in multiple oxidation states and structural environments such as spinel or layered phases, which significantly influence its leachability [20,21]. In some spent lithium-ion battery cathode materials, Mn dissolution is limited by structural stability or the formation of less soluble phases, making its recovery more sensitive to initial composition and acid availability than to operating conditions alone.

The importance of reducing agent concentration, particularly for Co (2.943) and Li (2.024), aligns with established redox-driven dissolution mechanisms. Transition metals in spent cathodes often exist in higher oxidation states such as Co^{3+} , which are less soluble under purely acidic conditions. The presence of a reducing agent facilitates conversion to more soluble lower oxidation states such as Co^{2+} , thereby enhancing dissolution efficiency. This mechanism has been widely reported in studies employing reductants such as hydrogen peroxide, organic acids, or alcohols, where improved recovery of Co and Ni is directly linked to reductive transformations within the cathode structure.

In the global averaged panel, leaching agent concentration ranked first (4.568) followed by temperature (3.401), with feed composition variables for Co, Mn, and Ni occupying the next tier (2.847, 2.637, and 2.355 respectively). This global ranking underscores that while process

conditions are the primary levers for overall system performance, material composition exerts a consistently significant and metal-specific influence that cannot be neglected in process optimisation. The SHAP analysis thus not only confirms established leaching principles but provides quantitative differentiation between process-driven and composition-driven recovery behaviour, highlighting the need for tailored leaching strategies for different cathode chemistries.

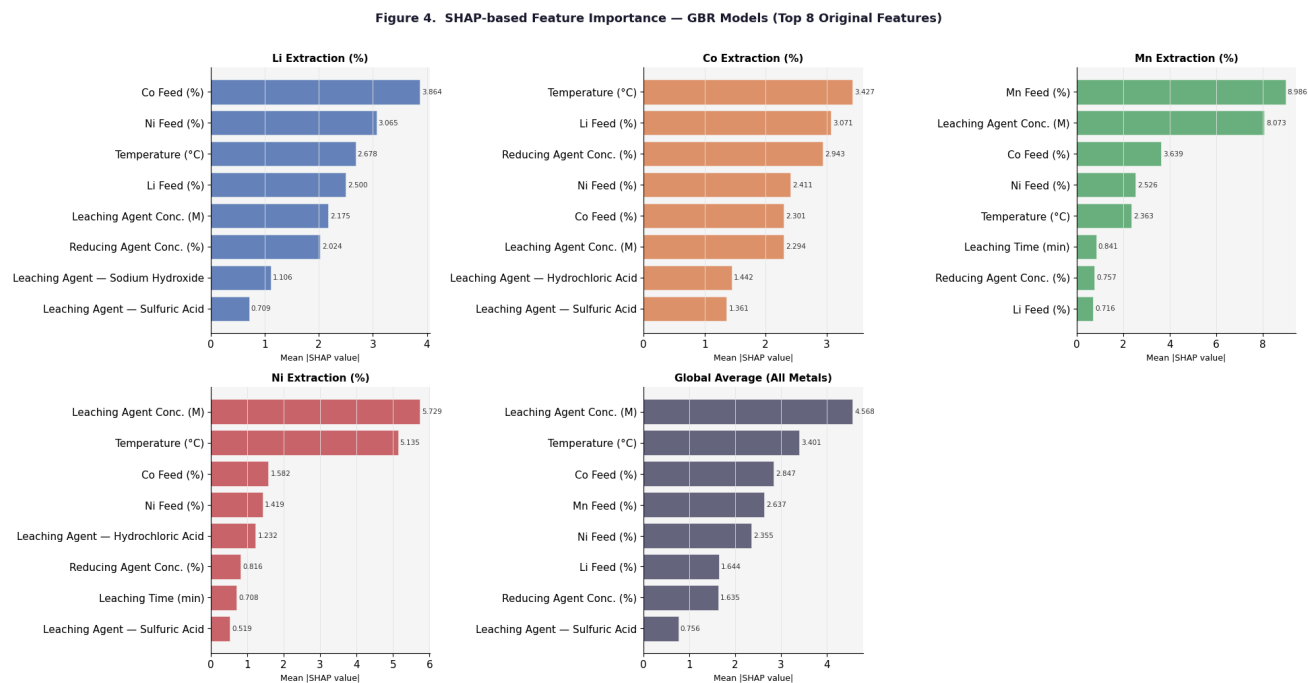


Figure 4. SHAP-based feature importance rankings

Table 3. Model hyperparameter values

ML Model	Hyperparameter	Value
Gradient Boosting Regression	n_estimators	300
	learning_rate	0.05
	max_depth	4
	subsample	0.8
	min_samples_split	2
	min_samples_leaf	5
	max_features	None
	loss	squared_error

The final optimized hyperparameters used for the GBR model across the leaching efficiency targets are detailed in Table 3. The configuration utilizes a moderate learning rate (`learning_rate` = 0.05) paired with 300 boosting stages (`n_estimators` = 300) which ensures a gradual learning process, allowing the model to sequentially correct residual errors from previous trees without hastily converging to sub-optimal local minima. Additionally, the individual decision trees are constrained to a relatively shallow maximum depth (`max_depth` = 4) to prevent the base learners from constructing overly complex rules that merely memorize noise in the training data. Lastly, by setting the minimum samples per leaf to 5 (`min_samples_leaf` = 5), model is required at least five experimental data points to form a terminal node, which ensures that its final predictions are representative of broader kinetic trends.

3.4 Model Performance Without Categorical Variables

Following SHAP-based feature importance analysis, the two categorical variables encoding leaching agent and reducing agent identity were removed to develop a more practically deployable model based solely on continuous process variables. The cross-validated baseline GBR model with categorical variables included achieved validation R^2 values of 0.8080, 0.7903, 0.7558, and 0.7613 for Li, Co, Mn, and Ni respectively on the original dataset.

Upon removing the categorical features, validation R^2 values shifted to 0.7731, 0.7529, 0.7613, and 0.7351 for Li, Co, Mn, and Ni respectively. These reductions range from approximately 0.003 to 0.026 across all targets, where the largest decline was observed for Nickel (from 0.7613 to 0.7351), while Manganese showed a marginal improvement (from 0.7558 to 0.7613), suggesting that categorical agent identity provides inconsistent and target-specific marginal benefit at small dataset scales. Test MAEs increased slightly across all metals, with Co rising from 8.09% to 8.78% and Ni from 7.16% to 7.79%, while Li and Mn remained within approximately 0.3–0.4 percentage points of their categorical-variable counterparts. Critically, these differences are well within the fold-to-fold standard deviation range reported for the original dataset, indicating the performance gap between the two feature configurations is not statistically meaningful at this data scale.

This pattern is consistent with the cross-validation analysis in Section 3.6, which showed that categorical variables provide negligible independent predictive signal when the dataset is small, because each leaching agent appears across too few varied conditions for the model to reliably learn agent-specific offsets. The continuous process variables, temperature, leaching agent concentration, and reducing agent concentration, are therefore sufficient for the GBR to capture the dominant thermodynamic and kinetic drivers of leaching behavior without requiring explicit reagent identity encoding. This numeric-only configuration offers meaningful practical advantages for process optimization and scale-up scenarios, where models must generalize across diverse leaching systems and reagent combinations without the constraint of categorical input requirements.

Table 4. R^2 , RMSE and MAE for the original dataset without categorical variables(after 5 folds)

Base Metrics	Model	Model	Train R ²	Train RMSE	Train MAE	Test R ²	Test RMSE	Test MAE
Li extracted (%)		GBR	0.9490	5.08	3.41	0.7731	10.60	6.39
Co extracted (%)		GBR	0.9194	7.64	5.11	0.7529	13.28	8.78
Mn extracted (%)		GBR	0.9502	6.11	3.35	0.7613	13.03	7.23
Ni extracted (%)		GBR	0.9323	6.47	3.93	0.7351	12.44	7.79

3.5 Bayesian Active Learning (BAL)

The effect of BAL augmentation on the training data distribution is illustrated through the box-and-whisker and violin plots (Figure 5). These plots compare the original and BAL-combined datasets across all eight continuous process variables. The BAL procedure identified the top 50% of data points with the highest prediction uncertainty, quantified as the standard deviation of predictions across the ensemble of weak learners within the Gradient Boosting Model (GBM), and concatenated these samples with the original dataset to form the augmented training set.

As shown in Figure 5, the BAL-augmented dataset exhibits a narrower interquartile range for several feed composition variables relative to the original dataset. The most pronounced reductions were observed in Mn Feed ($\Delta\text{IQR} = -0.618$) and Ni Feed ($\Delta\text{IQR} = -0.528$), followed by a moderate reduction in temperature ($\Delta\text{IQR} = -0.327$) and a minor reduction in Co Feed

($\Delta\text{IQR} = -0.069$). In contrast, leaching agent concentration (Conc. M), reducing agent concentration (Conc. %), and leaching time remained essentially unchanged ($\Delta\text{IQR} \approx 0.000$), indicating that the active learning step did not introduce additional spread in these variables. This pattern suggests that the GBM uncertainty was highest in regions where feed composition varied moderately, rather than in regions of extreme process conditions, and that the augmentation selectively concentrated sampling around the central range of those variables rather than expanding toward the tails.

The GBM-based uncertainty landscape reflects the sequential boosting structure of the model, where residual errors are concentrated in specific feature subspaces, leading BAL to reinforce coverage in compositionally intermediate regions rather than at the extremes.

The violin plots further illustrate the redistribution of sample density: variables that experienced IQR contraction (particularly Mn Feed and Ni Feed) show noticeably narrower violin bodies in the BAL dataset, with reduced spread at intermediate values. Variables where IQR was preserved, such as Conc. (M) and Time, display near-identical violin shapes between the two datasets, confirming that BAL did not alter the sampling density in those dimensions. The violin plots additionally reveal that the overall distributional shape is retained across all variables, with no evidence of multimodality being introduced by augmentation, suggesting that the BAL step added structurally consistent samples rather than introducing outlying or artifactual data points.

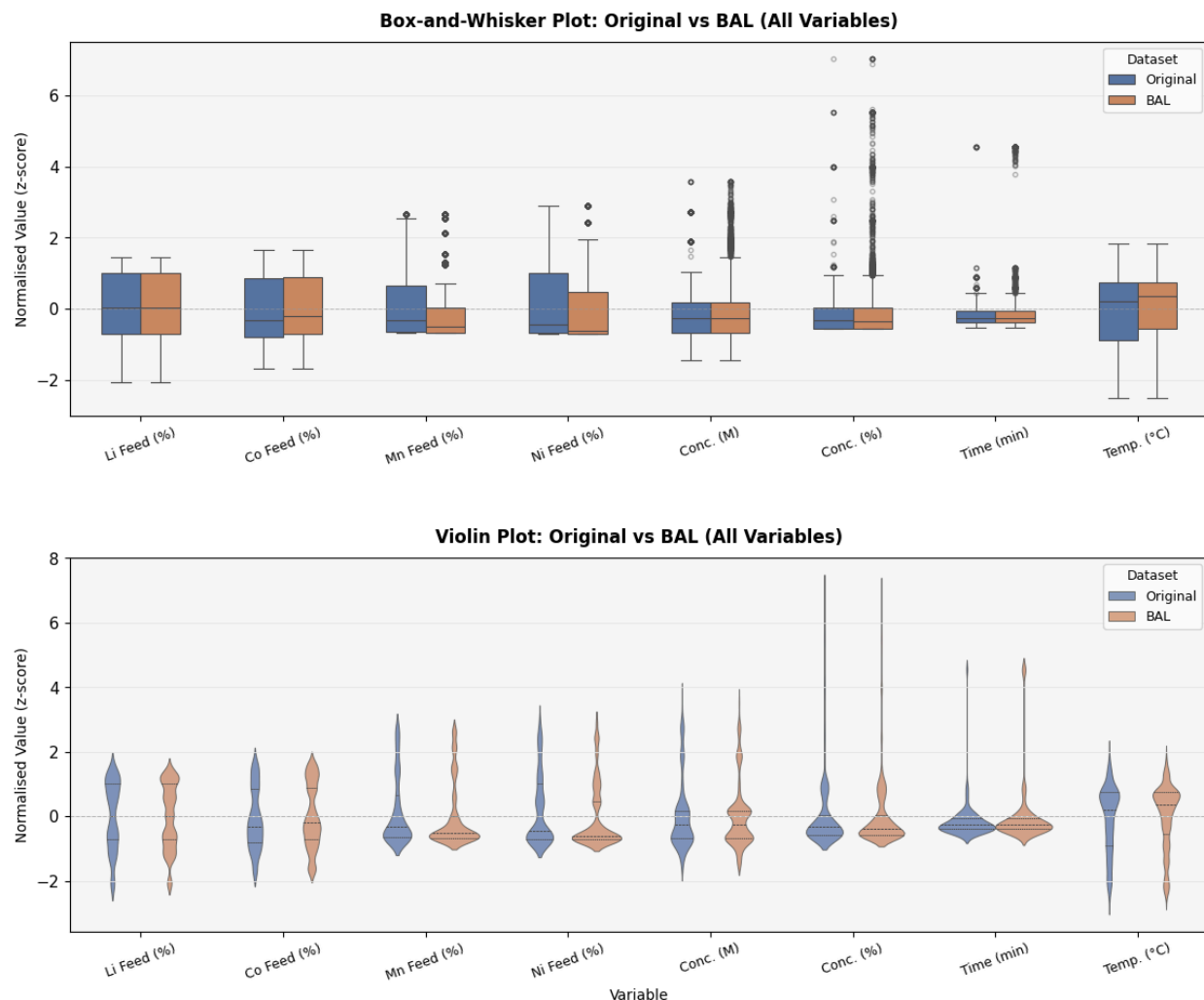


Figure 5. Box-and-whisker and Violin plots for BAL and original datasets

— IQR comparison (Original vs BAL) —

Li Feed (%)	IQR original=1.721	IQR BAL=1.721	$\Delta=+0.000$
Co Feed (%)	IQR original=1.672	IQR BAL=1.603	$\Delta=-0.069$
Mn Feed (%)	IQR original=1.305	IQR BAL=0.687	$\Delta=-0.618$
Ni Feed (%)	IQR original=1.695	IQR BAL=1.167	$\Delta=-0.528$
Conc. (M)	IQR original=0.856	IQR BAL=0.856	$\Delta=+0.000$
Conc. (%)	IQR original=0.609	IQR BAL=0.611	$\Delta=+0.002$
Time (min)	IQR original=0.330	IQR BAL=0.330	$\Delta=+0.000$

Temp. (°C)

IQR original=1.637 IQR BAL=1.310 $\Delta=-0.327$

The box-and-whisker plots highlight the shift in the interquartile range and median values across key process variables between the original and BAL-augmented datasets, providing a concise summary of how the active learning step redistributed sample density. While these plots capture the central tendency and spread of each variable effectively, they do not fully convey the shape of the underlying distributions, particularly in cases where data is asymmetric or concentrated near a narrow range of values. The violin plots complement this view by overlaying kernel density estimates on the same data, revealing finer structural differences in distribution shape between the two datasets that are not visible from the box plots alone.

3.6 Final Model Performance Evaluation

Five-fold cross-validation was conducted across four model configurations arising from two dataset variants (Original and Combined) and two feature set conditions (with and without categorical variables). For each configuration, per-metal and pooled performance metrics such as R^2 , RMSE, and MAE were collected and reported separately for training folds (Table A) and validation folds (Table B), with variability expressed as the standard deviation across the five folds. The training-fold metrics characterise in-sample fit consistency, while the validation-fold metrics provide the primary measure of generalisation to held-out data (see below).

Table 5. Test and train data for the final forward models with both BAL and original data (with categorical variables). Standard deviation was calculated using 5-fold Cross-Validation Results

Original dataset

Table. A 5-fold CV — Training folds (mean $\pm$ std)

target	Train					
	R^2		RSME		MAE	
	Original	Standard deviation	Original	Standard deviation	Original	Standard deviation
Li	0.9521	0.0016	4.93	0.09	3.40	0.07

Co	0.9360	0.0021	6.81	0.12	4.71	0.08
Mn	0.9535	0.0098	5.90	0.63	3.19	0.26
Ni	0.9469	0.0088	5.72	0.43	3.47	0.21
All (pooled)	0.9471	0.0096	5.84	0.77	3.69	0.62

Table B. 5-fold CV — Validation folds (mean $\pm$ std)

	Test					
	R ²		RSME		MAE	
	Original	Standard deviation	Original	Standard deviation	Original	Standard deviation
Li	0.8080	0.0177	9.80	1.07	6.07	0.63
Co	0.7903	0.0360	12.23	1.37	8.09	0.86
Mn	0.7558	0.0943	13.19	2.25	7.20	0.37
Ni	0.7613	0.0837	11.78	1.14	7.16	0.32
All(Pooled)	0.7789	0.0695	11.75	1.97	7.13	0.92

Results for the combined dataset

Table. A 5-fold CV — Training folds (mean $\pm$ std)

	Train		
	R ²	RSME	MAE

	Train					
	Combined	Standard deviation	Combined	Standard deviation	Combined	Standard deviation
Li	0.9614	0.0008	4.06	0.04	2.76	0.02
Co	0.9295	0.0012	6.25	0.06	4.41	0.04
Mn	0.9739	0.0023	4.31	0.18	2.60	0.07
Ni	0.9520	0.0031	4.88	0.17	3.37	0.10
All (pooled)	0.9542	0.0164	4.88	0.86	3.28	0.71

Table B.5-fold CV — Validation folds (mean $\pm$ std)

	Test					
	R ²		RSME		MAE	
target	Combined	Standard deviation	Combined	Standard deviation	Combined	Standard deviation
Li	0.9455	0.0057	4.80	0.10	3.21	0.06
Co	0.9046	0.0061	7.26	0.18	5.12	0.10
Mn	0.9286	0.0203	7.01	0.82	4.12	0.48
Ni	0.9008	0.0154	6.98	0.50	4.71	0.26
All (pooled)	0.9199	0.0226	6.51	1.11	4.29	0.77

Table 6. Test and train data for the final forward models with both BAL and original data (without categorical variables). Standard deviation was calculated using 5-fold Cross-Validation Results

Results for model without categorical fields

Original dataset

Table. A 5-fold CV — Training folds (mean $\pm$ std)

	Train					
	R ²		RSME		MAE	
	Original	Standard deviation	Original	Standard deviation	Original	Standard deviation
Li	0.9490	0.0049	5.08	0.24	3.41	0.14
Co	0.9194	0.0022	7.64	0.10	5.11	0.06
Mn	0.9502	0.0098	6.11	0.60	3.35	0.26
Ni	0.9323	0.0093	6.47	0.36	3.93	0.19
All (pooled)	0.9377	0.0146	6.33	0.99	3.95	0.73

Table B. 5-fold CV — Validation folds (mean $\pm$ std)

	Test					
	R ²		RSME		MAE	
	Original	Standard deviation	Original	Standard deviation	Original	Standard deviation
Li	0.7731	0.0313	10.60	0.92	6.39	0.40

Co	0.7529	0.0387	13.28	1.35	8.78	0.85
Mn	0.7613	0.0932	13.03	2.18	7.23	0.44
Ni	0.7351	0.0839	12.44	1.02	7.79	0.27
All(Pooled)	0.7556	0.0689	12.34	1.79	7.55	1.02

Results for the combined dataset

Table. A 5-fold CV — Training folds (mean $\pm$ std)

target	Train					
	R ²		RSME		MAE	
	Combined	Standard deviation	Combined	Standard deviation	Combined	Standard deviation
Li	0.9552	0.0009	4.37	0.04	2.96	0.03
Co	0.9163	0.0015	6.81	0.04	4.79	0.03
Mn	0.9722	0.0024	4.45	0.18	2.71	0.07
Ni	0.9432	0.0031	5.31	0.15	3.66	0.09
All (pooled)	0.9467	0.0205	5.24	0.99	3.53	0.81

Table B.5-fold CV — Validation folds (mean $\pm$ std)

	Test		
	R ²	RSME	MAE

target	Test					
	Combined	Standard deviation	Combined	Standard deviation	Combined	Standard deviation
Li	0.9335	0.0085	5.30	0.23	3.51	0.14
Co	0.8854	0.0064	7.96	0.16	5.63	0.11
Mn	0.9263	0.0209	7.12	0.84	4.28	0.46
Ni	0.8875	0.0166	7.43	0.48	5.12	0.24
All (pooled)	0.9082	0.0262	6.95	1.12	4.64	0.85

3.6.1 Overall Analysis

Across all four configurations, training-fold R^2 values were uniformly high, ranging from approximately 0.92 to 0.97, and were accompanied by low standard deviations (typically < 0.01 to 0.02) which might imply that model that converges reliably regardless of which 80% of data is used for training. The GBR architecture, with its sequential residual-correction mechanism, is well-suited to the non-linear dependencies present in leaching systems, where temperature, acid concentration, and time interact in multiplicative rather than additive ways.

In the Original dataset configurations, pooled validation R^2 falls to approximately 0.78 (with categoricals) and 0.76 (without categoricals), compared to training R^2 values near 0.95 and 0.94 respectively. This is a drop of approximately 17 percentage points in both cases, where the gap can be the signature of mild overfitting, which is expected when a flexible ensemble model is trained on a relatively small and potentially sparse dataset. The engineered interaction features (Time $\times$ Temperature, Leaching Concentration $\times$ Time, and polynomial terms) increase the effective dimensionality of the input space, meaning with limited training data, the model partially memorises the training distribution rather than learning robust generalisable relationships. Notably, the magnitude of this gap is similar whether categorical variables are

included or not in the Original dataset, suggesting that the overfitting is primarily driven by dataset size rather than by the encoding of leaching agent identity.

On the other hand, the combined dataset shows that pooled validation R^2 rises to 0.92 (with categoricals) and 0.91 (without categoricals), compared to training R^2 values of approximately 0.95 in both cases with a gap of only 3–4 percentage points. This is the single most important finding across all eight tables. The additional data introduced in the Combined dataset substantially reduces the variance of the learned decision boundaries, allowing the model to generalise rather than interpolate memorised training patterns, hence fix for the bottleneck in the data volume.

3.6.2 Effect of Categorical Variables

In the Original dataset, the with-categorical and without-categorical configurations produce almost identical validation R^2 values (0.779 vs. 0.756, pooled), which might imply that categorical identity adds limited independent predictive information when the dataset is small. However, when the combined dataset is introduced, the with-categorical model achieves a pooled validation R^2 of 0.920 compared to 0.908 for the without-categorical model, and validation MAE of 4.29% versus 4.64%. This improvement is consistent across all four metal targets and indicates that, given sufficient data to anchor the per-agent estimates, leaching agent identity contributes genuine predictive signal. Different leaching agents (e.g., sulfuric acid, citric acid, hydrogen peroxide as reducer) operate through distinct electrochemical mechanisms and have different selectivities for Li, Co, Mn, and Ni. When the dataset is large enough that each agent appears across a range of conditions, the model can learn these agent-specific offsets reliably. The implication for future modelling efforts is that categorical process variables should be retained and that further data collection targeted at underrepresented agent types would likely yield continued gains.

3.6.3 Per-Metal Predictability

Lithium (Li) achieves the highest and most stable validation R^2 in every configuration, ranging from 0.808 (Original, with categoricals) to 0.946 (Combined, with categoricals), with the lowest standard deviations across folds. This stability is likely connected to the Lithium leaching efficiency being high across a wide range of conditions and being less sensitive to fine variations

in reducing agent concentration than the transition metals. The relative homogeneity of Lithium recovery values may make this target intrinsically easier to predict with fewer data points, as the signal-to-noise ratio in the target variable is higher.

Cobalt (Co) on the other hand consistently produces the second-highest validation R^2 in Combined dataset configurations (0.905 with categoricals, 0.885 without) but shows the largest fold-to-fold standard deviation of any metal in the Original dataset validation (R^2 std = 0.036 with categoricals). This sensitivity might be connected to the sparse coverage in the original data. Co leaching is strongly dependent on the oxidation state of the cathode material and on the concentration of the reducing agent, making it sensitive to local gaps in the experimental design. The improvement seen in the Combined dataset (std dropping to 0.006 with categoricals) confirms that these gaps are being filled.

Thirdly, Manganese (Mn) shows the most pronounced fold-to-fold instability in the Original dataset, with validation R^2 standard deviations of 0.094 (with categoricals) and 0.093 (without) which is nearly an order of magnitude larger than the corresponding training standard deviations (0.010 and 0.010). This extreme variability indicates that Mn is highly dependent on which samples are held out, where some folds likely contain conditions (particularly high-temperature or high-acid-concentration regimes) where the training data is too sparse to support reliable extrapolation. Mn dissolution chemistry is more complex than Li or Co in certain leaching systems, with dissolution rates depending non-linearly on pH and exhibiting different behaviour in reductive versus oxidative leaching environments. In the Combined dataset, Mn validation R^2 stabilises markedly (std = 0.020 with categoricals, 0.021 without), and mean performance rises to 0.929, suggesting that the additional data covers the previously sparse regions of the Mn-relevant operating space more completely.

In the same way, Nickel (Ni) mirrors Mn in exhibiting elevated fold-to-fold variance in the Original dataset (validation R^2 std = 0.084 and 0.084 for the two configurations), however, unlike Mn, which achieves a validation R^2 of 0.929 in the best Combined dataset configuration, Ni's best validation R^2 is 0.901 which is the lowest among the four metals in the Combined dataset. Ni leaching efficiency in mixed-cathode systems is known to be particularly sensitive to the stoichiometry and morphology of the precursor material, which are not directly captured by the current numeric and categorical features. This might imply that additional features such as

cathode phase composition, particle size distribution, or solid-to-liquid ratio might be required to close the remaining variance.

3.6.4 Uncertainty and Fold Stability as a Diagnostic

In the Original dataset, large standard deviations (for example Mn validation RMSE std = 2.25% for Original with categoricals) indicate that reported mean metrics have high uncertainty. This is because the Original dataset metrics characterise average performance across splits rather than providing tight bounds on expected performance on new data.

In the Combined dataset, standard deviations contract substantially across all metals and both feature configurations. For the pooled validation set, RMSE standard deviation falls from approximately 1.97% (Original with categoricals) to 1.11% (Combined with categoricals), and MAE standard deviation falls from 0.92% to 0.77%. This contraction reflects a more uniform data distribution, where any five-fold partition produces folds of comparable difficulty, resulting in the Combined dataset performance estimates being substantially more reliable.

3.6.5 Summary and final interpretations

In conclusion, the cross-validation results establish the following 3 results below

First, dataset volume is the primary determinant of generalisation performance in this modelling task. Moving from the Original to the Combined dataset improves pooled validation R^2 by approximately 14 percentage points regardless of whether categorical features are included. Second, categorical variables (leaching agent type and reducing agent type) provide consistent, physically grounded predictive value, but only when sufficient data are available to estimate per-agent effects reliably; their benefit is negligible at small sample sizes and becomes meaningful at the scale of the Combined dataset.

Third, per-metal predictability follows the order $\text{Li} > \text{Co} > \text{Mn} \approx \text{Ni}$, with Mn and Ni showing the greatest sensitivity to data coverage gaps. This might arise from undersampled combinations of agent type, temperature, and concentration for these two targets.

Since

Partial Dependence Plots

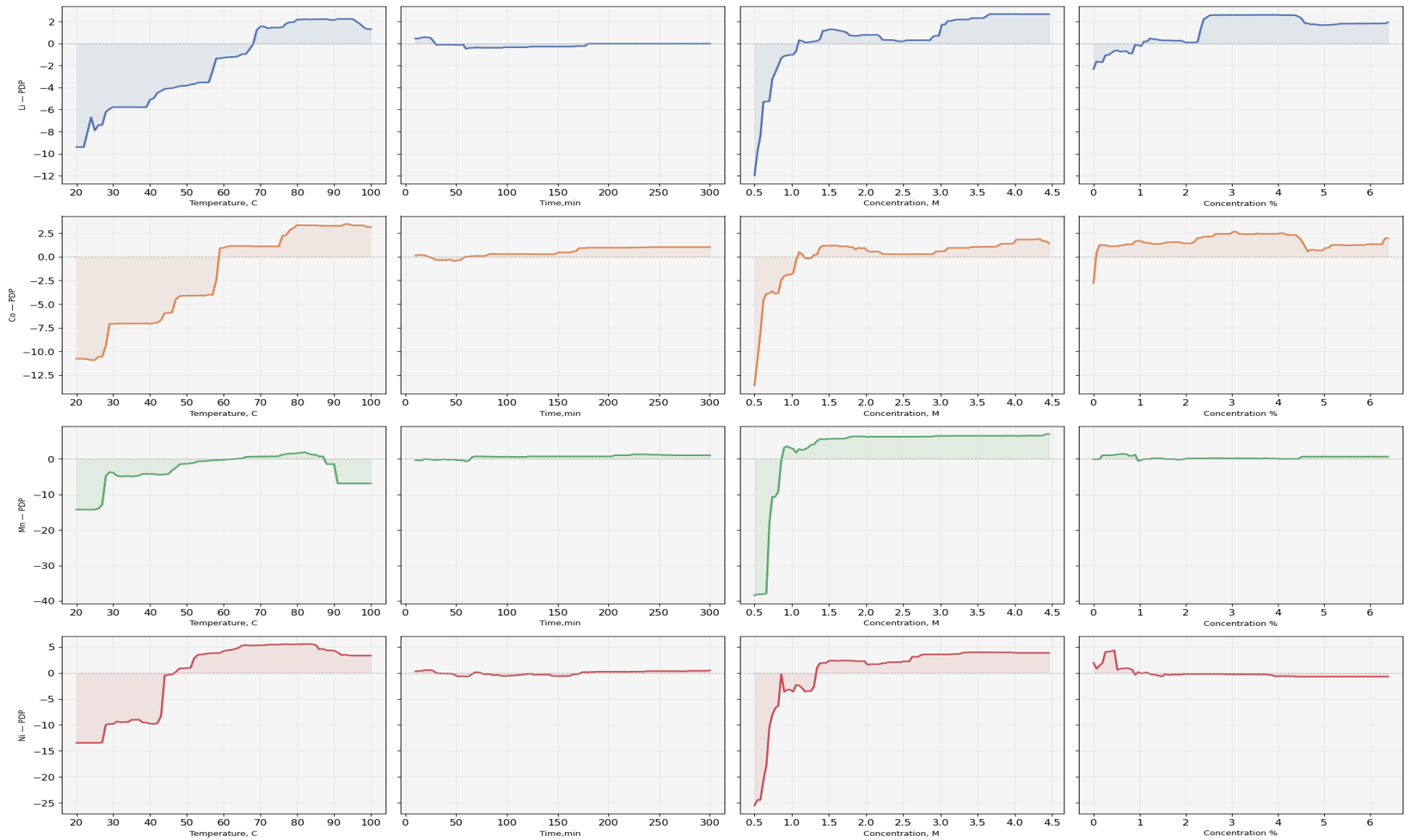


Figure 6: PDP showing the influence of several input variables on Li and Co recovery.

The partial dependence plots (PDPs) for all four target metals reveal consistent and physically interpretable trends across the key process variables.

For Li recovery, temperature exerts the strongest influence, with a steep monotonic increase from strongly negative PDP values at low temperatures ($\sim 20^{\circ}\text{C}$) to a plateau near 2 percentage points above the mean at temperatures exceeding 60°C , indicating that thermal energy is critical for dissolving Li-bearing phases. Leaching agent concentration (M) similarly drives a sharp positive transition between approximately 0.5 and 1.5 M before leveling off, suggesting a threshold acidity requirement. Time and reducing agent concentration (%) show comparatively modest and largely flat effects across their respective ranges, implying that once minimum leaching conditions are met, further increases in these variables contribute marginally to Li recovery.

For Co recovery, the response patterns closely mirror those of Li. Temperature produces a pronounced step-wise increase with the most rapid gain occurring between 20°C and 60°C , after which the PDP stabilises. Leaching agent concentration follows a similarly sharp sigmoidal rise, plateauing above approximately 1.5 M. Time and reducing agent concentration again contribute minimally, with near-flat PDP profiles across their full ranges, suggesting Co dissolution is governed primarily by temperature and acid/reagent strength rather than reaction duration or reductant availability.

For Mn recovery, the concentration dependence is notably more abrupt than for the other metals. The PDP for leaching agent concentration (M) shows a steep, near-vertical rise centered around 0.5–1.0 M, transitioning from strongly negative (approximately -35 percentage points) to slightly positive values within a very narrow concentration window. This threshold behaviour suggests that Mn-bearing phases have a sharply defined dissolution boundary and are highly sensitive to reagent concentration. Temperature also produces a negative-to-positive transition, though with a more gradual profile and a local minimum near $60\text{--}80^{\circ}\text{C}$ before partially recovering, which may reflect competing dissolution and re-precipitation mechanisms. Time and reducing agent concentration show minimal influence, consistent with the pattern observed for Li and Co.

For Ni recovery, leaching agent concentration (M) again produces the most prominent effect, with a sharp positive inflection between 0.5 and 1.0 M followed by a plateau, indicating that sufficient acidity or alkalinity is critical for dissolving Ni-bearing phases. Temperature shows a

clear monotonic positive effect, improving recovery progressively across the full range examined. In contrast to the other metals, the reducing agent concentration (%) for Ni exhibits a slight initial positive response at very low values before flattening, and notably shows a small negative dip at intermediate concentrations (~1–2%), suggesting that excess reductant may not benefit and could slightly interfere with Ni dissolution. Time remains largely non-influential across all four metals, indicating that the leaching reactions approach equilibrium relatively rapidly under the conditions represented in the dataset.

Across all metals, near-flat PDP profiles for time and reducing agent concentration across Li, Co, Mn, and Ni suggest that within the ranges captured in the compiled dataset, these variables operate past their rate-limiting thresholds for most experimental conditions, and that optimisation efforts should prioritise reagent concentration and thermal conditions.

Since the model with categorical variables consistently outperforms the configuration without them, it has been established as the final model and will be used exclusively for all further analysis and comparisons. Further data visualizations for each fold runs can be found in the appendix below

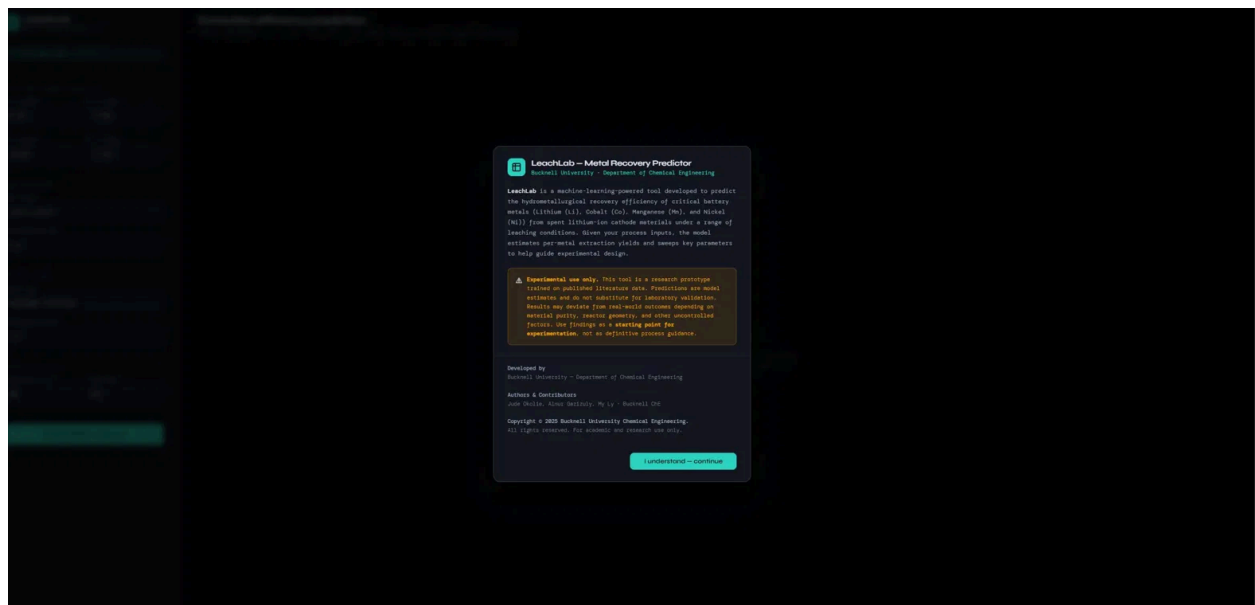
3.7 GUI Interface

To make the final forward model accessible without requiring programming expertise, an interactive web application LeachLab: Metal Recovery Predictor was developed using Render and deployed at <https://leaching-app-1.onrender.com/>. Upon launch, users are presented with a disclaimer modal acknowledging the research-prototype nature of the tool, noting that predictions are model estimates trained on published literature data and should be used as a starting point for experimental design rather than as definitive process guidance.

The input panel allows users to specify the eight continuous process variables that includes Li, Co, Mn, and Ni feed compositions (%), leaching agent type and concentration (M), reducing agent type and concentration (%), reaction temperature (°C), and leaching time (min) and based

on the inputs provided, receive predicted leaching efficiencies for all four metals simultaneously. The results dashboard presents per-metal recovery estimates as individual metric cards, a comparative bar chart, and a radar plot displaying the recovery profile across all four metals, enabling rapid visual assessment of relative recovery balance under the specified conditions.

A parameter sensitivity sweep section sweeps each process variable across its physically relevant range while all other inputs are held fixed at the user-entered values, with all sweep values representing real model predictions rather than interpolated or simulated curves. A model verdict panel at the base of the dashboard automatically identifies single-variable optima for each swept parameter and quantifies the predicted gain in average recovery relative to the current input, providing actionable guidance for experimental condition selection.



LeachLab

Metals Recovery Predictor

Global GEM model - active

GLOBAL FEED COMPOSITION (%)

Li In Feed

6.79

Co In Feed

17.68

Mn In Feed

16.46

Ni In Feed

17.58

LEACHING AGENT

Agent type

Lactic Acid

Concentration (%)

1.5

REDUCING AGENT

Agent type

Hydrogen Peroxide

Concentration (%)

1.5

PROCESS CONDITIONS

Temperature (°C)

60

Time (min)

50

Run predictions + sweep

Extraction efficiency prediction

Global GEM model - Li - Co - Mn - Ni - all sweep values are real model predictions

No prediction yet

Enter your experimental conditions and press Run. Sweep events call the GEM model - no simulated or make-up values.

LeachLab

Metals Recovery Predictor

Global GEM model - active

GLOBAL FEED COMPOSITION (%)

Li In Feed

6.79

Co In Feed

17.68

Mn In Feed

16.46

Ni In Feed

17.58

LEACHING AGENT

Agent type

Lactic Acid

Concentration (%)

1.5

REDUCING AGENT

Agent type

Hydrogen Peroxide

Concentration (%)

1.5

PROCESS CONDITIONS

Temperature (°C)

60

Time (min)

50

Run prediction + sweep

Extraction efficiency prediction

Global GEM model - Li - Co - Mn - Ni - all sweep values are real model predictions

Global Li

87.3%

Global Co

85.5%

Global Mn

80.3%

Global Ni

78.3%

PREDICTED EFFICIENCY BY METAL

100%

90%

80%

70%

60%

50%

40%

30%

20%

10%

0%

Li

Co

Mn

Ni

RECOVERY PROFILE - ALL METALS

Li

Co

Mn

Ni

EXPERIMENTAL CONDITIONS LOG

LEACHING AGENT

Lactic Acid

REDUCING AGENT

Hydrogen Peroxide

Concentration (%)

1.5

Concentration (%)

1.5

Temperature (°C)

60

Time (min)

50

Global Li

87.3%

Global Co

85.5%

Global Mn

80.3%

Global Ni

78.3%

PARAMETER SENSITIVITY SWEEP

REAL MODEL PREDICTIONS - NO SIMULATION

TEMPERATURE

0.001 - 0.002 - 0.003 - 0.004 - 0.005 - 0.006 - 0.007 - 0.008 - 0.009 - 0.010 - 0.011 - 0.012 - 0.013 - 0.014 - 0.015 - 0.016 - 0.017 - 0.018 - 0.019 - 0.020

0.001

0.002

0.003

0.004

0.005

0.006

0.007

0.008

0.009

0.010

0.011

0.012

0.013

0.014

0.015

0.016

0.017

0.018

0.019

0.020

LEACHING CONCENTRATION

0.001 - 0.002 - 0.003 - 0.004 - 0.005 - 0.006 - 0.007 - 0.008 - 0.009 - 0.010 - 0.011 - 0.012 - 0.013 - 0.014 - 0.015 - 0.016 - 0.017 - 0.018 - 0.019 - 0.020

0.001

0.002

0.003

0.004

0.005

0.006

0.007

0.008

0.009

0.010

0.011

0.012

0.013

0.014

0.015

0.016

0.017

0.018

0.019

0.020

REACTION TIME

0.001 - 0.002 - 0.003 - 0.004 - 0.005 - 0.006 - 0.007 - 0.008 - 0.009 - 0.010 - 0.011 - 0.012 - 0.013 - 0.014 - 0.015 - 0.016 - 0.017 - 0.018 - 0.019 - 0.020

0.001

0.002

0.003

0.004

0.005

0.006

0.007

0.008

0.009

0.010

0.011

0.012

0.013

0.014

0.015

0.016

0.017

0.018

0.019

0.020

REDUCING AGENT CONCENTRATION

0.001 - 0.002 - 0.003 - 0.004 - 0.005 - 0.006 - 0.007 - 0.008 - 0.009 - 0.010 - 0.011 - 0.012 - 0.013 - 0.014 - 0.015 - 0.016 - 0.017 - 0.018 - 0.019 - 0.020

0.001

0.002

0.003

0.004

0.005

0.006

0.007

0.008

0.009

0.010

0.011

0.012

0.013

0.014

0.015

0.016

0.017

0.018

0.019

0.020

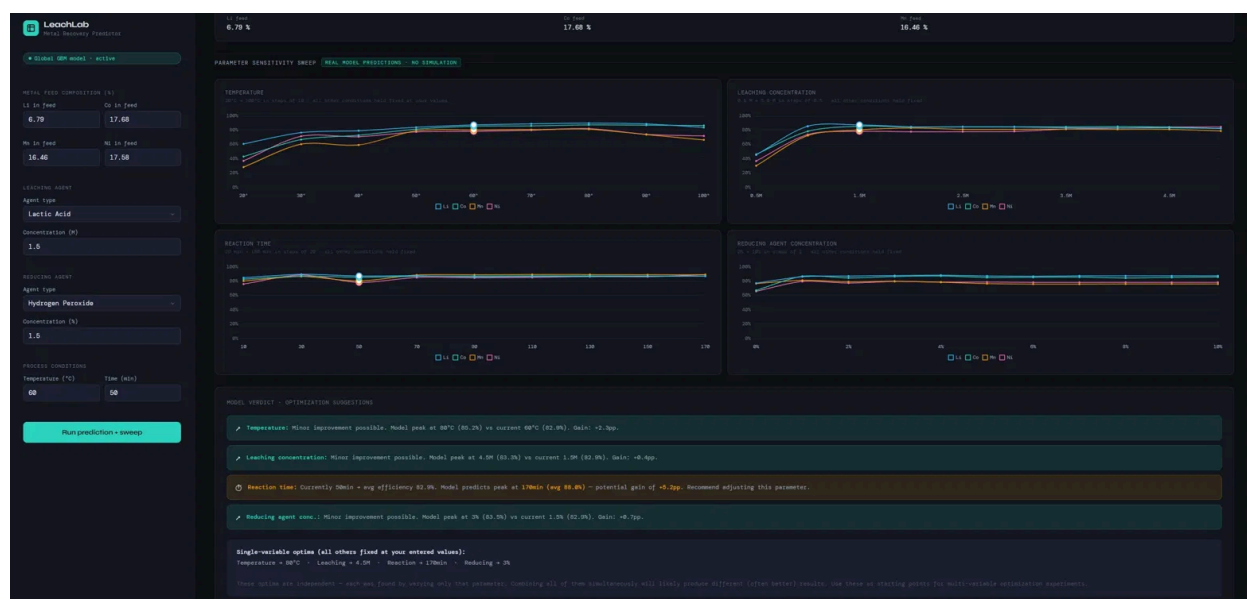


Figure 8. GUI Interface

3.8 Model validation with Experimental Results

Leaching experiments were performed and the results were compared with model predictions. The procedures for all leaching experiments including each efficiency calculations is shown in Figure 9 while figures 10 – 11 shows the leaching experiment comparison results for Li, Co and Mn respectively. Note all experiments were performed at 60 mins with the use of organic acid and hydrogen peroxide as a reductant.

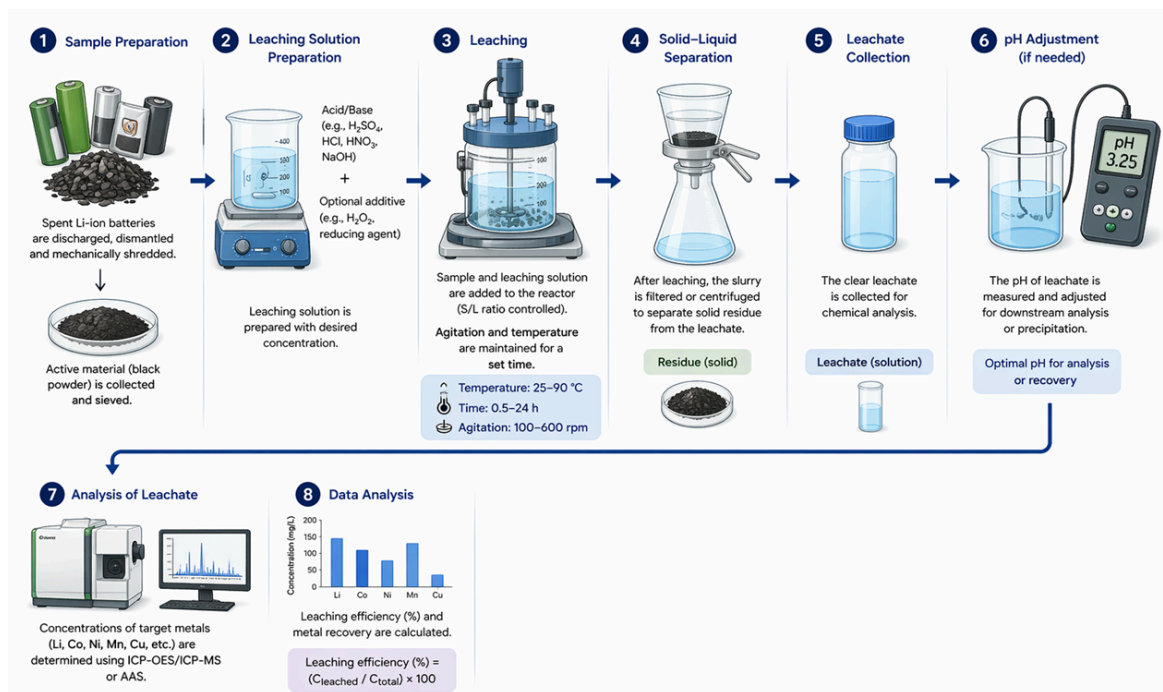


Figure 9. Schematics of the leaching experimental procedure

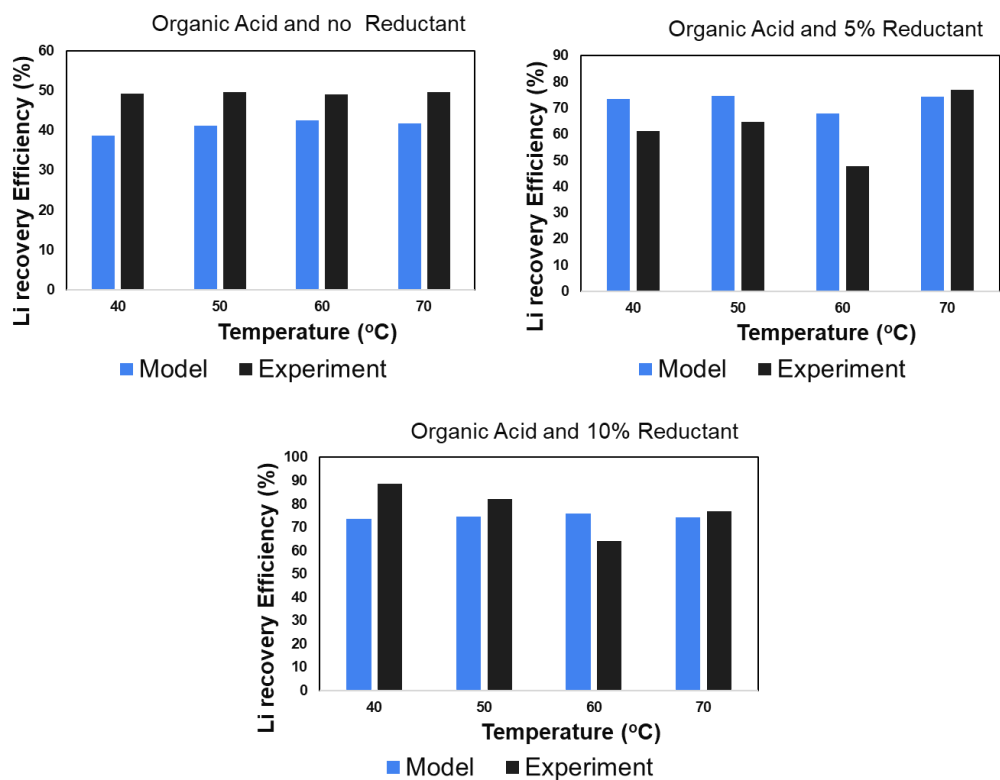


Figure 10. Comparison of Li recovery efficiency between experimental and model results

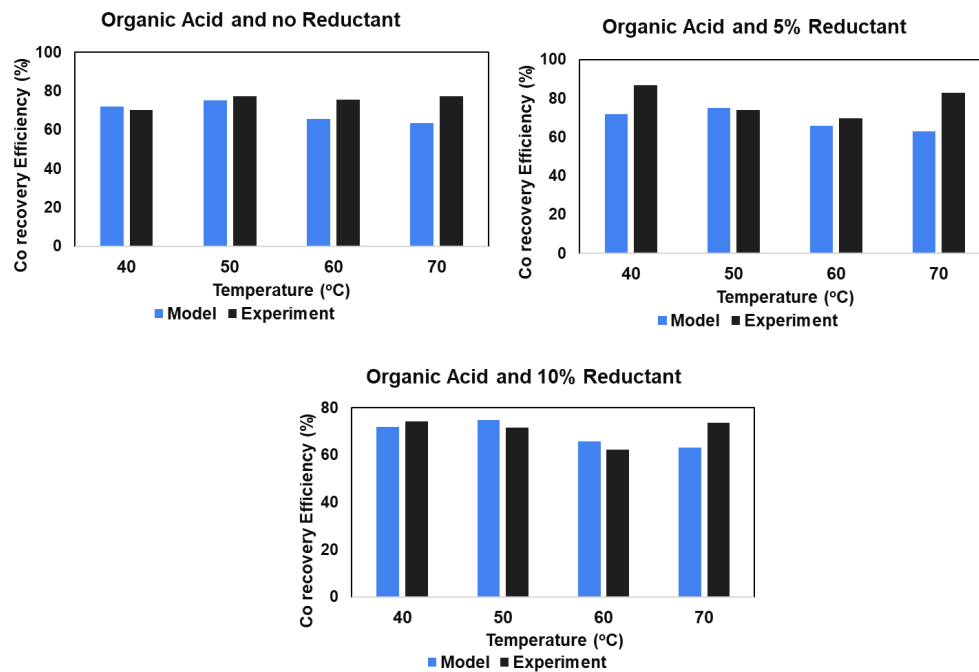


Figure 11. Comparison of Co recovery efficiency between experimental and model results

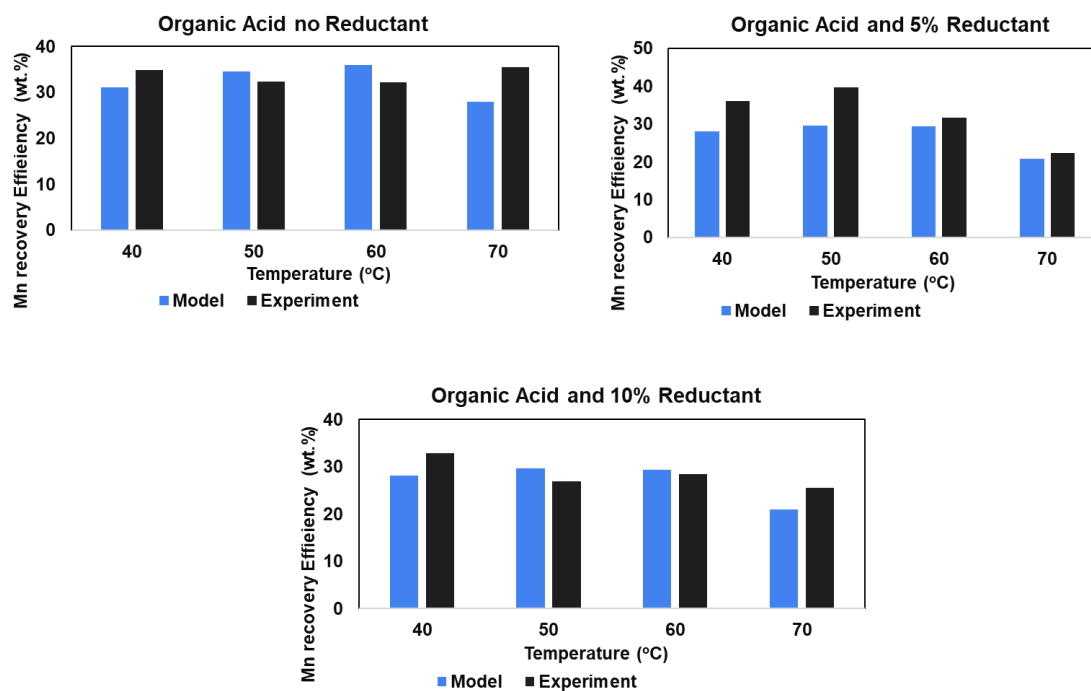


Figure 12. Comparison of Mn recovery efficiency between experimental and model results

Table 7. Two-sample t-test p-values comparing model predictions and experimental results.

Variable	p value
Li	0.9041
Co	0.9746
Mn	0.6705
Ni	0.6680

3.8 Model Validation with Experimental Results

Leaching experiments were performed across four temperature levels (40, 50, 60, and 70°C) under three reductant conditions (no reductant, 5% hydrogen peroxide, and 10% hydrogen peroxide) using organic acid as the leaching agent, with all experiments conducted at a fixed reaction time of 60 minutes. The resulting Li, Co, and Mn recovery efficiencies were compared directly against model predictions to assess the forward model's ability to generalize to independently generated experimental data. The experimental procedures and efficiency calculations are illustrated in Figure 9, while Figures 10, 11, and 12 present the bar chart comparisons for Li, Co, and Mn recovery respectively.

Under the no-reductant condition, both model and experiment consistently predict Li recovery in the 38 to 50% range across temperatures, with the model slightly underpredicting experimental values by approximately 8 to 10 percentage points at most temperatures. When 5% hydrogen peroxide is added, recovery increases substantially for both model and experiment into the 60 to 80% range, though the model tends to overpredict at lower temperatures (40 and 50°C) while converging more closely with experimental values at 70°C. At 10% reductant concentration, the agreement between model and experiment is tightest, with both falling in the 65 to 88% range and deviations generally remaining within 10 percentage points. The overall pattern confirms that the model correctly identifies the positive contribution of reductant addition to Li recovery, even if it does not perfectly resolve the magnitude of that effect at every condition.

For cobalt recovery (Figure 11), model and experimental results show considerably closer agreement across all three reductant conditions compared to lithium. Under the no-reductant condition, predicted and measured Co recoveries track each other tightly in the 63 to 78% range across all temperatures, with most bar pairs differing by less than 5 percentage points. At 5% hydrogen peroxide, the model slightly underpredicts at 40°C where the experiment reaches approximately 85% while the model predicts near 68%, but agreement improves at 50, 60, and 70°C. Under 10% reductant conditions, the correspondence is again strong, with both model and experiment remaining within the 60 to 75% range and no systematic directional bias apparent across the temperature range examined. The relatively flat temperature response observed for Co in both the experimental and predicted data is consistent with the PDP analysis in Section 3.6, which showed that Co recovery plateaus above approximately 60°C, and with the cross-validation results that identified Co as the second most predictable target after Li.

For manganese recovery (Figure 12), the recovery values are considerably lower than those for Li and Co, consistent with the known structural stability of Mn-bearing oxide phases under mild organic acid leaching conditions. Across all three reductant conditions, both model predictions and experimental measurements fall predominantly within the 20 to 40 wt.% range. Under the no-reductant condition, model and experiment track closely across the 40 to 60°C range, with deviations of approximately 3 to 5 percentage points, though a mild divergence appears at 70°C where the model underpredicts the experimental recovery. The 5% reductant condition shows slightly larger scatter between model and experiment, particularly at 40 and 50°C where experimental values reach approximately 35 to 40 wt.% against model predictions near 27 to 29 wt.%. At 10% reductant, the agreement is again reasonable, with both sets of values converging in the 20 to 32 wt.% range across all temperatures. The residual deviations observed for Mn are consistent with the higher fold-to-fold variance reported for Mn in the cross-validation analysis, reflecting the inherent difficulty of predicting Mn dissolution under conditions where cathode structural effects and reagent availability interact non-linearly.

To statistically assess whether model predictions and experimental measurements were drawn from the same underlying distribution, a two-sample t-test was performed for each target metal. The resulting p-values are reported in Table 7. All four metals yielded high p-values, 0.9041 for

Li, 0.9746 for Co, 0.6705 for Mn, and 0.6680 for Ni, confirming that there is no statistically significant difference between the two sets of values at the 95% confidence level. Users may enter the leaching agent as a broad category such as organic acid or inorganic acid, or as a specific reagent identity, and model performance remains consistent across both input conventions. This flexibility reflects the finding from Section 3.3 that leaching agent type contributes only moderately to predictive power relative to continuous process variables, and that the dominant drivers of metal recovery are captured robustly by the numeric feature set. Taken together, the experimental validation and statistical testing confirm that the final model provides reliable and generalized predictions of leaching efficiency across a practical range of operating conditions.

4. Conclusions

This study presented a machine learning framework for predicting hydrometallurgical leaching efficiencies of lithium, cobalt, manganese, and nickel from spent lithium-ion battery cathode materials. A compiled dataset of 992 entries drawn from peer-reviewed literature was used to develop and evaluate a Gradient Boosting Regressor (GBR) pipeline across four model configurations, varying dataset composition (original versus BAL-augmented combined) and feature set (with and without categorical variables). The GBR consistently delivered strong predictive performance, achieving test R^2 values of 0.9473, 0.9132, 0.9528, and 0.9286 for Li, Co, Mn, and Ni respectively on the original dataset with categorical variables included.

SHAP-based feature importance analysis revealed that leaching agent concentration and temperature were the dominant global predictors of metal recovery, with mean absolute SHAP values of 4.568 and 3.401 respectively across all targets. Feed composition variables, particularly Co Feed (%) and Mn Feed (%), exerted metal-specific secondary influence, especially on Li and Mn dissolution. Notably, the removal of categorical variables (leaching agent and reducing agent identity) produced only marginal reductions in test R^2 of approximately 0.003 to 0.02 across all four metals, confirming that the continuous numeric feature set captures the dominant physicochemical drivers of leaching behavior and that the model generalizes well without reliance on reagent identity encoding.

A synthetic data augmentation procedure inspired by Bayesian Active Learning principles was implemented to address data sparsity in underrepresented regions of the process space. Moving from the original to the combined augmented dataset improved pooled cross-validation R^2 by approximately 14 percentage points, from 0.779 to 0.920 with categorical variables included, and reduced fold-to-fold RMSE standard deviation from 1.97% to 1.11%. These gains were consistent across all four metals and both feature configurations, demonstrating that dataset volume rather than model architecture was the primary bottleneck governing generalization performance. Per-metal predictability followed the order $\text{Li} > \text{Co} > \text{Mn} \approx \text{Ni}$, with Mn and Ni exhibiting the greatest sensitivity to data coverage gaps, reflected in their elevated fold-to-fold variance on the original dataset.

Partial dependence plots provided showed that temperature and leaching agent concentration each produced steep, threshold-type transitions in predicted recovery across all four metals, consistent with established hydrometallurgical kinetic principles. Leaching time and reducing agent concentration exhibited comparatively flat PDP profiles across their full ranges, suggesting that these variables operate past their rate-limiting thresholds for the majority of experimental conditions represented in the compiled dataset. The sharp concentration-dependent dissolution boundary observed specifically for Mn is consistent with the known structural complexity of Mn-bearing oxide phases and their sensitivity to reagent availability.

Model predictions were validated against independently conducted laboratory leaching experiments using organic acid and hydrogen peroxide as reductant across temperatures of 40–70°C and reductant concentrations of 0%, 5%, and 10%. Two-sample t-tests comparing model predictions to experimental outcomes yielded p-values of 0.9041, 0.9746, 0.6705, and 0.6680 for Li, Co, Mn, and Ni respectively, confirming no statistically significant difference between predicted and measured recovery values at the 95% confidence level. The final model was deployed as an interactive web application, LeachLab, providing accessible, real-time leaching efficiency predictions and parameter sensitivity analysis without requiring programming expertise.

Future work should prioritize expanding the training dataset with experimentally validated entries targeting undersampled conditions, particularly for Mn and Ni dissolution across diverse cathode chemistries and agent combinations. Incorporating additional physicochemical

descriptors such as cathode phase composition, particle size distribution, and solid-to-liquid ratio may further close the residual predictive variance observed for Ni. Extending the pipeline to include a reverse optimization model capable of recommending process conditions for a user-specified recovery target would complete the framework and substantially enhance its value for practical process design in spent lithium-ion battery recycling.

5. Declaration of AI Use

AI was used to write and debug code written for the machine learning model presented in this work. All results generated by AI were reviewed by the author. The author takes full responsibility for the accuracy and credibility of this work.

Github Repository

The code and further research paper details can be found at:

<https://github.com/AlnurGazizuly/Leaching-App>

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Appendix A

Further data visualization insights on model training

Original dataset with categorical fields

Actual vs Predicted Leaching Efficiency

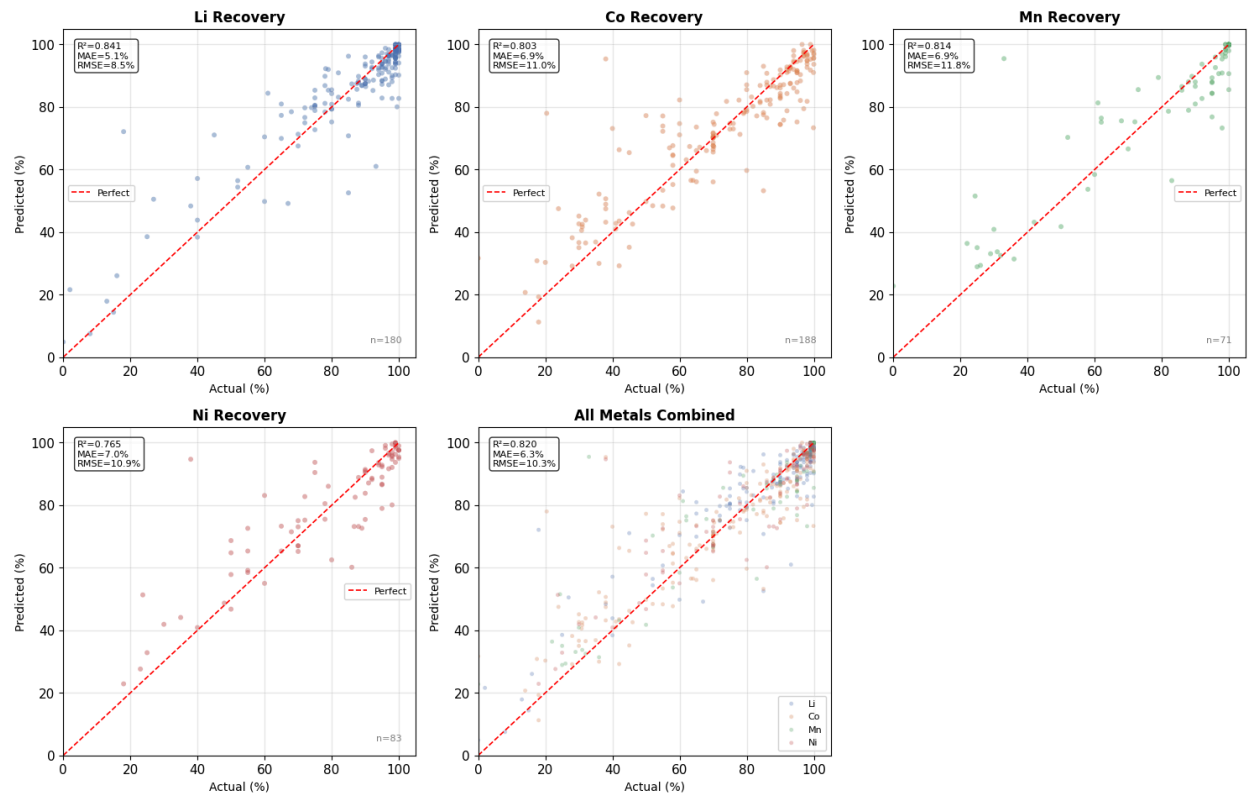
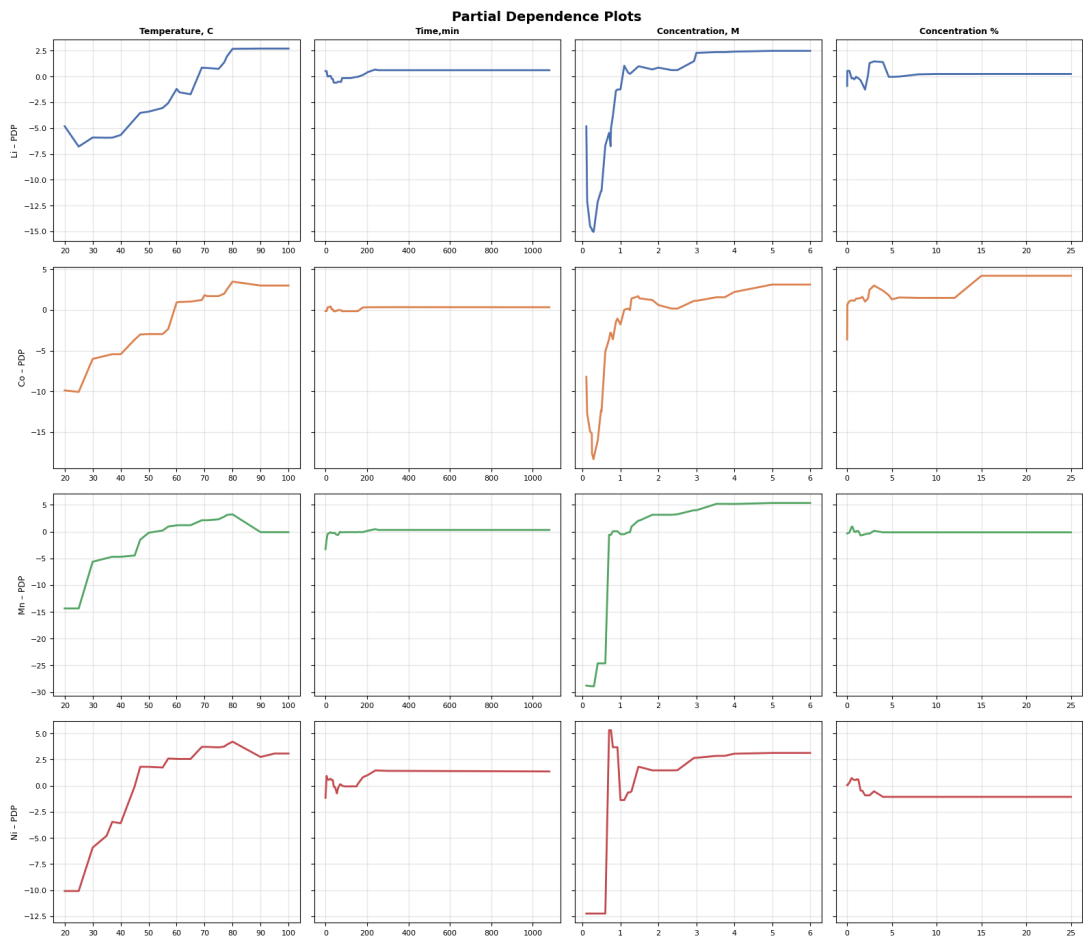
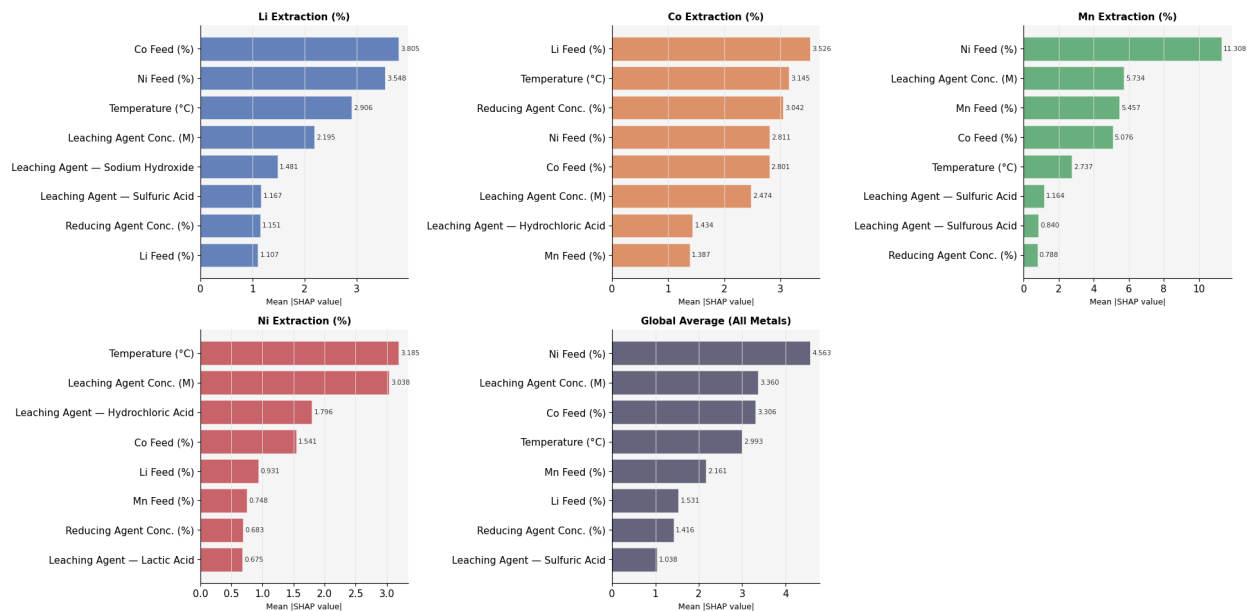


Figure 4. SHAP-based Feature Importance — GBR Models (Top 8 Original Features)



Original dataset without categoricals.

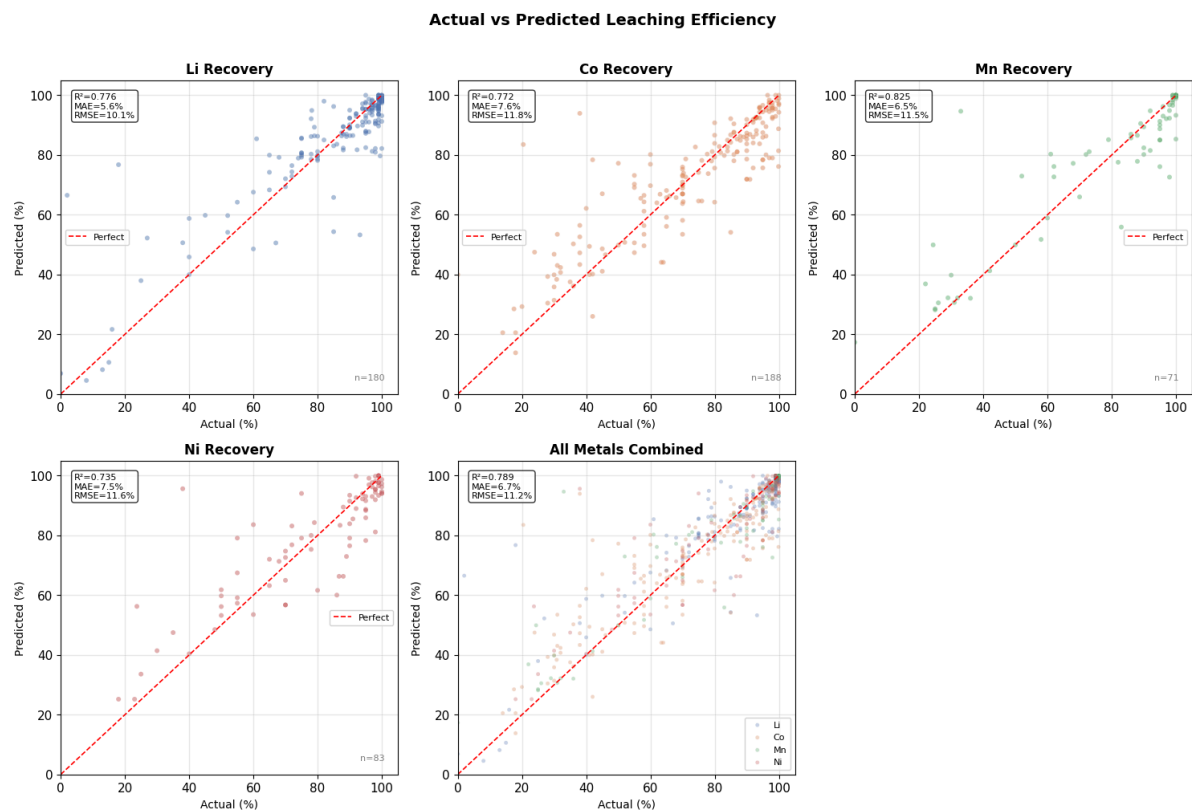
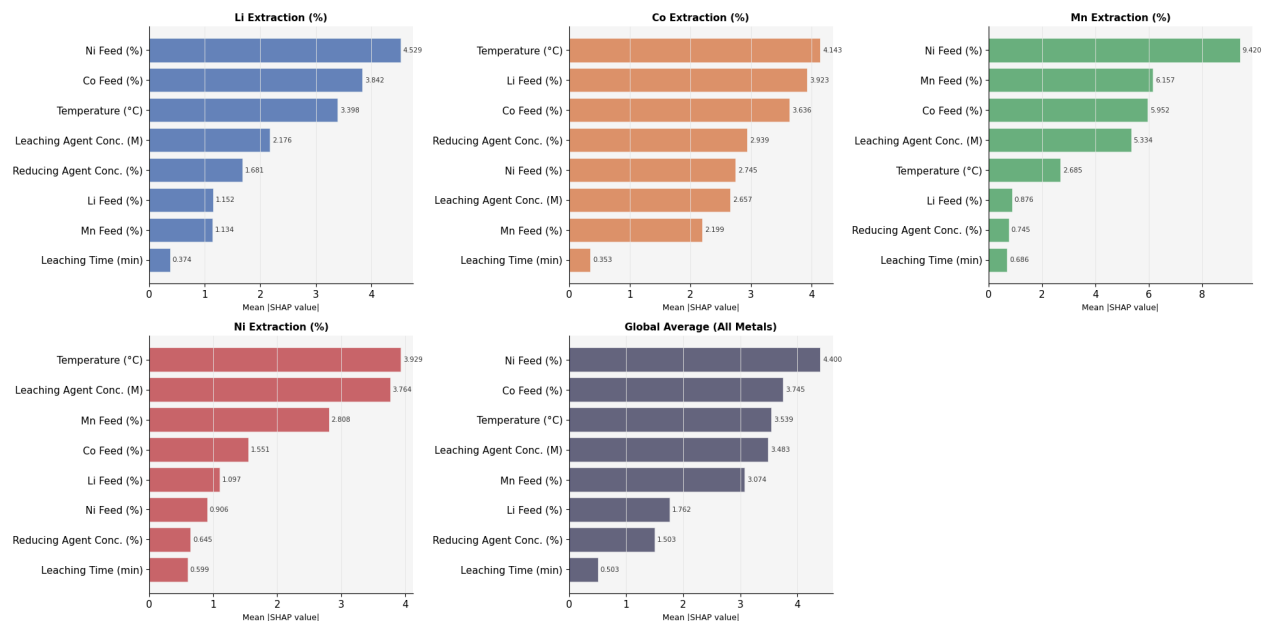
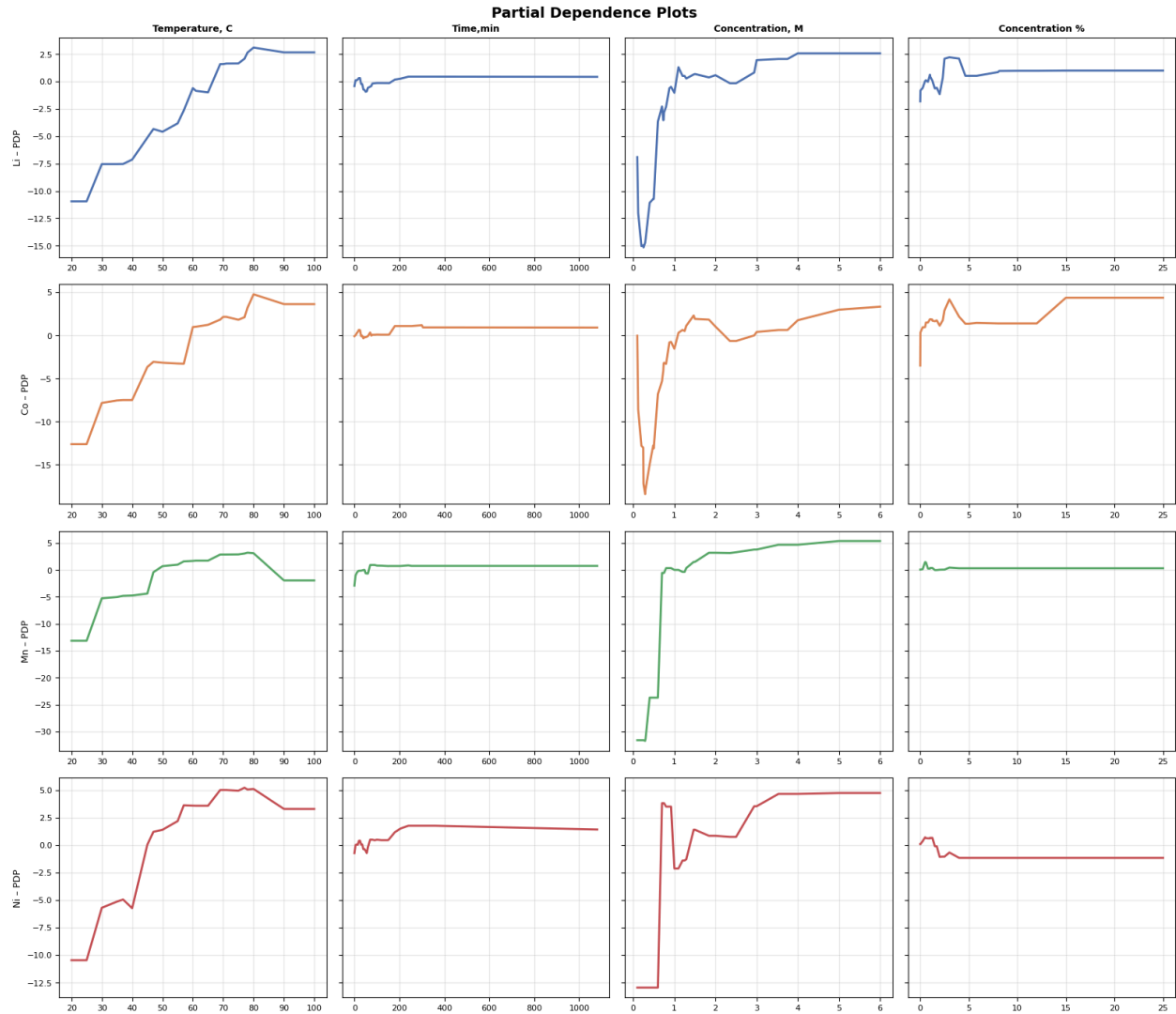


Figure 4. SHAP-based Feature Importance — GBR Models (Top 8 Original Features)





Combined dataset with categoricals

Actual vs Predicted Leaching Efficiency

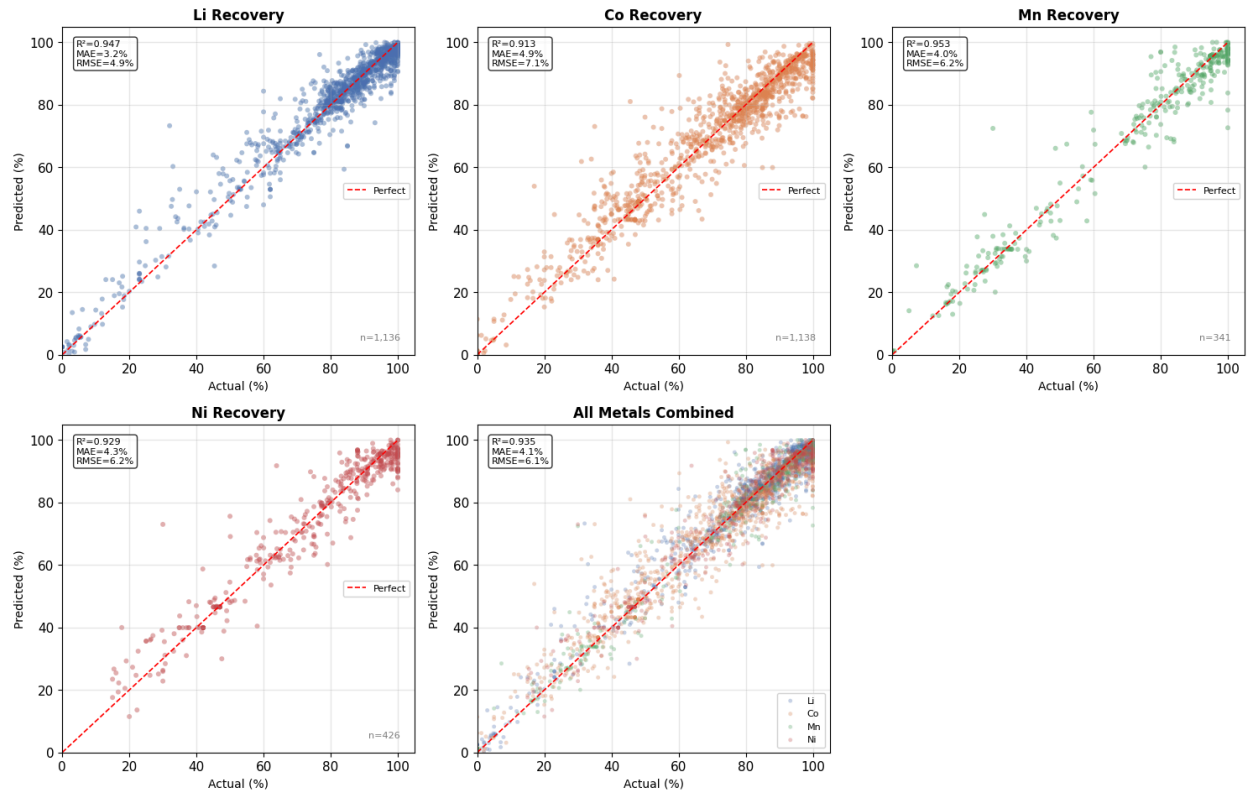
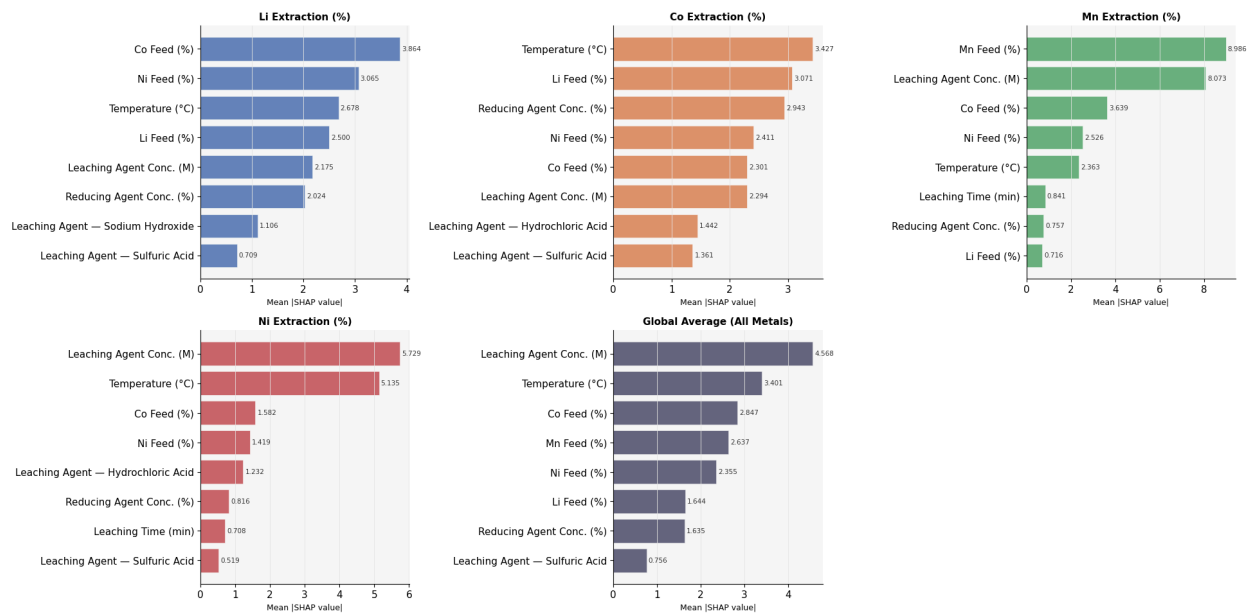
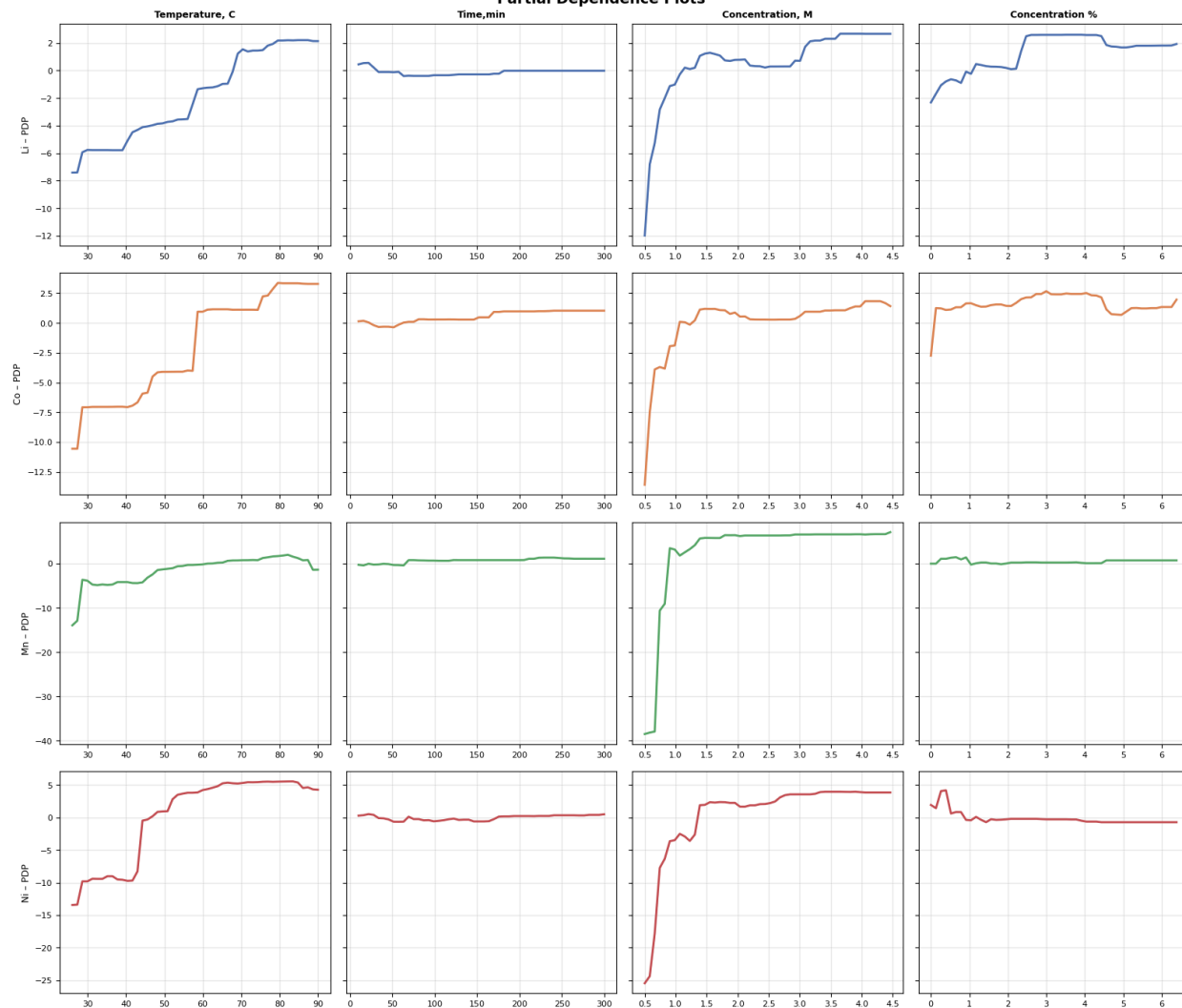


Figure 4. SHAP-based Feature Importance — GBR Models (Top 8 Original Features)



Partial Dependence Plots



Combined dataset without categoricals

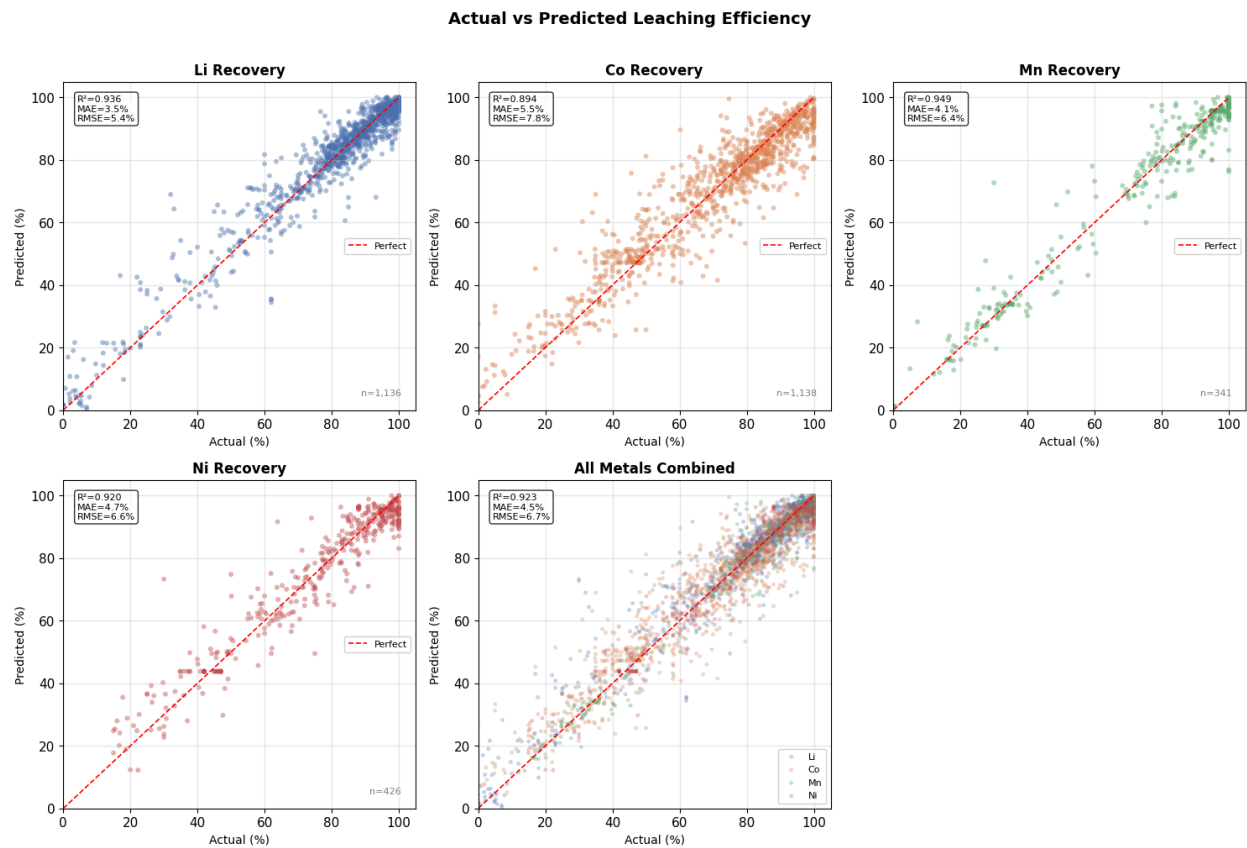
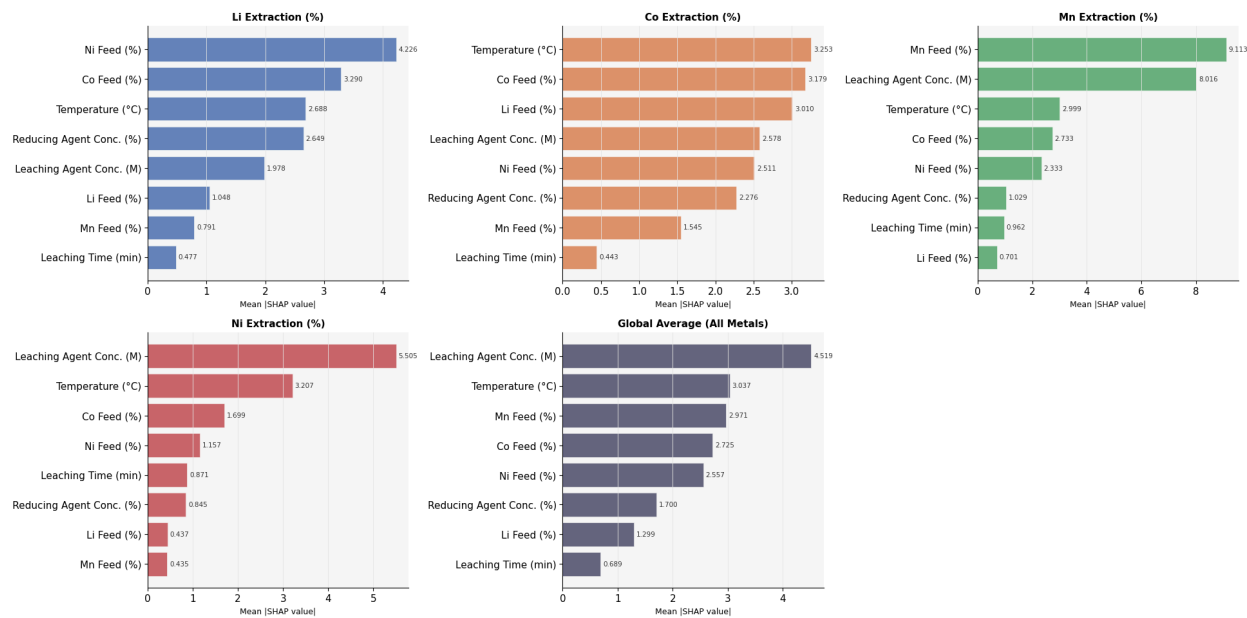


Figure 4. SHAP-based Feature Importance — GBR Models (Top 8 Original Features)



Partial Dependence Plots

